



Research on dynamic graph isomorphism network for tool wear stage monitoring based on multi-source information fusion

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ABSTRACT

Tool wear is a prominent unfavorable factor that directly affects the machining accuracy and efficiency of CNC machine tools. Accurate classification monitoring of tool wear stage is of great significance for high performance machining. Existing monitoring methods based on deep learning rarely improves the accuracy of identification by establishing non Euclidean structural relationships between multi-sensor signals. A new method for tool wear stage classification monitoring based on graph neural network (GNN) is proposed to address these limitations. Using convolutional neural network (CNN) and gramian angular summation field (GASF) to extract spatiotemporal features from samples. Subsequently, by taking into account both the static and dynamic characteristics of temporal data, the time series is transformed into a graph structure using cosine similarity. Following this, a graph isomorphism network (GIN) is employed to deeply integrate information from neighboring nodes. The dimensionality of the feature matrix and adjacency matrix is progressively reduced through temporal graph pooling (TGP). Finally, the classification monitoring of three stages of tool wear is realized. The experimental validation was conducted based on the public PHM2010 dataset. The results show that the proposed model achieves an average accuracy of 96.51 % in the state recognition task, improved by 0.97 % over the current state-of-the-art model. In addition, the recognition accuracy of multi-source information fusion is significantly higher than single-signal. The ablation experiments further validate the positive contribution of each module to the monitoring performance. In summary, the study provides an efficient and robust solution for tool wear monitoring.

1. Introduction

As an important participant in the machining process, cutting tools will inevitably suffer from wear due to a variety of reasons, such as mechanical action, heat, and chemical factors, which would directly affect the surface quality and productivity of parts, and in extreme cases may also lead to safety risks [1,2]. Therefore, in order to reasonably control the replacement time of tool and improve economic efficiency, it is necessary to study the accurate monitoring of tool wear state. Usually, the tool wear process throughout the life cycle is generally divided into the initial wear stage, normal wear stage and severe wear stage [3,4]. In order to ensure the dimensional accuracy and surface roughness of finished parts, the severe wear stage should be avoided, and the tool should be changed in a timely manner before the arrival of severe stage. At the same time, the premature replacement will waste the tool life and

increase manufacturing costs. During the actual machining process, the control of tool wear mainly relies on passive organs or manual experience, which is very easy to cause misjudgment and omission. In recent years, in order to give full play to the tool cutting performance, a majority of scholars have conducted extensive and in-depth research on tool wear state monitoring [5–7].

Different from the tool wear monitoring that quantitatively analyzes the amount of wear width value on the flank face, the tool wear stage identification aims to establish a nonlinear mapping relationship between the signal characteristics and wear state, and then quickly classify the wear stage. Generally speaking, the tool wear stage recognition has been mainly categorized into direct and indirect methods [8]. However, due to many limitations in the application of direct methods, most researchers prefer to use indirect methods [9–11]. Indirect method refers to the identification of wear state from real-time signals, that is acquired

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for constructing a correlation model between the tool wear state and monitored data. Compared with the direct method, the indirect method can realize continuous and real-time monitoring without affecting the machining process [12]. Since signals such as cutting force [13,14], acoustic emission [15], vibration [16], current and power [17,18] are closely related to the tool wear condition, they are often used for on-line monitoring of tool wear state.

Tool wear is a smooth variation process, in the transition region of different wear stages, it is difficult to distinguish the tool wear due to the blurred boundaries [19]. The training efficiency and recognition accuracy of tool wear classification model depend on feature extraction. The traditional feature extraction methods involve extracting features from the time domain, frequency domain, and time–frequency domain. Sultan Binsaeid et al. [20] extracted different time and frequency domain features from sensor signals, such as force, vibration, acoustic emission, and spindle power. They used a data acquisition and signal processing module and a correlation-based feature selection technique to evaluate the importance of these features. Liao et al. [21] employed time domain statistics, frequency domain analysis, and wavelet packet decomposition to extract features from cutting force signals, and further utilized a genetic algorithm to identify the sensitive features that are highly correlated with the tool wear state. Based on the extracted sensitive features, many machine learning methods have been widely applied to monitor the tool wear status. The commonly used machine learning models include: artificial neural networks (ANN) [22], support vector machines (SVM) [23], decision trees (DT) [24], linear discriminant analysis (LDA) [25], k-nearest neighbors (KNN) [26], and hidden Markov models (HMD) [27].

Despite the success of machine-learning based tool wear state recognition models, they are usually built using shallow layers and it is difficult to learn deep features from large amounts of data. At the same time, the manual extraction process may lead to the loss of more important information features in the original signal [28]. With the further development of machine learning, deep learning (DL) can automatically extract more advanced features from the original signals to realize “end-to-end” tool wear state recognition [29–32]. Wang et al. [33] proposed a denoising transformer autoencoder for preprocessing of tool condition monitoring (TCM) classifiers, which enhances the extraction of wear features through a reconstruction task. Tao et al. [34] developed an integrated model with long-short-term memory (LSTM) network and HMM for tool wear state classification in milling machining. Considering that CNN are efficient and flexible in image processing, some researchers have converted one-dimensional time-series signals into two-dimensional images, and then used the channel-space attention mechanism to improve the accuracy of feature extraction for tool wear information, which can reduce the model misclassification [35]. Wei et al. [36] used continuous wavelet transform to convert the force and vibration signals into time–frequency images, and realized multi-source information fusion through CNN, and finally introduced the channel-space attention module to adaptively adjust the weights. Dong et al. [37] converted one-dimensional force signals into two-dimensional images based on Markov transition field, and then realized tool wear classification using the adaptive-visual-Transformer-Kolmogorov-Arnold network (AViTK). However, the aforementioned research has primarily focused on sensor feature extraction methods within Euclidean spaces. The exploration of potential non-Euclidean structural relationships in tool wear monitoring signals remains limited, indicating a research gap in this area.

Sensor signals can be modeled as graphs, each feature can be represented as a node, interconnected through hidden dependencies, thereby establishing non-Euclidean structural relationships. GNN captures the dependencies of graph by passing messages between nodes. In recent years, GNN variants such as graph convolutional networks (GCN), graph attention networks (GAT), graph recurrent networks (GRN), and GIN have demonstrated breakthrough performance in many deep learning tasks [38]. For example, GNN have achieved remarkable

success with graph-structured data such as social networks and medicine [39]. In the field of industrial monitoring, Li et al. [40] constructed a GNN-based intelligent fault diagnosis and prediction framework, effectively addressing the modeling challenge of irregular graph data in non-Euclidean spaces. Wang et al. [41] innovatively proposed a dynamic spatiotemporal graph model integrating performance degradation and fault propagation. By incorporating unsupervised learning mechanisms, it compensates for existing methods’ shortcomings in leveraging engineering prior knowledge. Zhou et al. [42] proposed the Hierarchical Temporal-Modal Synchronization Fusion Framework (HTMSFF), leveraging GNN technology to effectively enhance tool wear recognition accuracy under multimodal monitoring data. Wu et al. [43] designed a multi-view fully connected GNN that extracts spatial dependencies through local and global graph convolutions, successfully achieving equipment remaining life prediction. Zhou et al. [44] developed a tool wear recognition model based on multi-scale edge, which utilizes labeled GNN for feature extraction on small samples. The proposed model outperforms CNN, AlexNet, and ResNet. In addition, Liu et al. [45] proposed the TGP method, which combines graph pooling with temporal processing, not only reducing the feature dimension but also retaining key temporal features. However, none of the studies in the field of tool wear monitoring have explored the GIN model, which demonstrates superior performance in image classification tasks.

In summary, many scholars have conducted extensive research on tool wear state recognition methods, but the author believes that there are still the following problems to be solved:

- (1) The existing models perform well on the specific datasets, while drop significantly in scenarios with large differences in data distribution, indicating that the generalization ability of the models still needs to be improved.
- (2) Most studies only rely on a single signal for tool wear state recognition, failing to fully utilize the complementary nature of multi-source information, which limits the further improvement in recognition accuracy.
- (3) Existing methods primarily focus on Euclidean spaces, rarely leveraging the potential non-Euclidean structural relationships inherent in temporal data, and insufficiently exploring the application of powerful GIN models.

In order to solve the above problems, this paper proposes a new method for online identification of tool wear stages based on GIN. The main innovations of this paper are as follows:

- (1) A new method for end-to-end tool wear stage classification monitoring based on GIN is proposed, and a time–space correlation model of multi-source sequence signals is established. By feature extraction on cutting force and vibration signals in time and space dimension, multi-source information fusion is realized, avoiding the time cost and professional dependence of manual feature extraction.
- (2) Model the time-series signal as a graph. By comprehensively considering the long-term stability and short-term time-varying nature of the temporal characteristics, static and dynamic graphs are constructed. Leveraging the non-Euclidean structural relationships inherent in the time-series signal enhances the accuracy of identifying tool wear states.
- (3) The GIN-based TGP method is introduced. This approach integrates graph pooling and temporal processing techniques to achieve feature dimensionality reduction while preserving critical temporal characteristics, making it particularly suitable for handling time-dependent graph data.

The rest of the paper is organized as follows: Section 2 presents the monitoring method. Section 3 describes the related experiments and collected data. Section 4 performs the experimental validation. Finally,

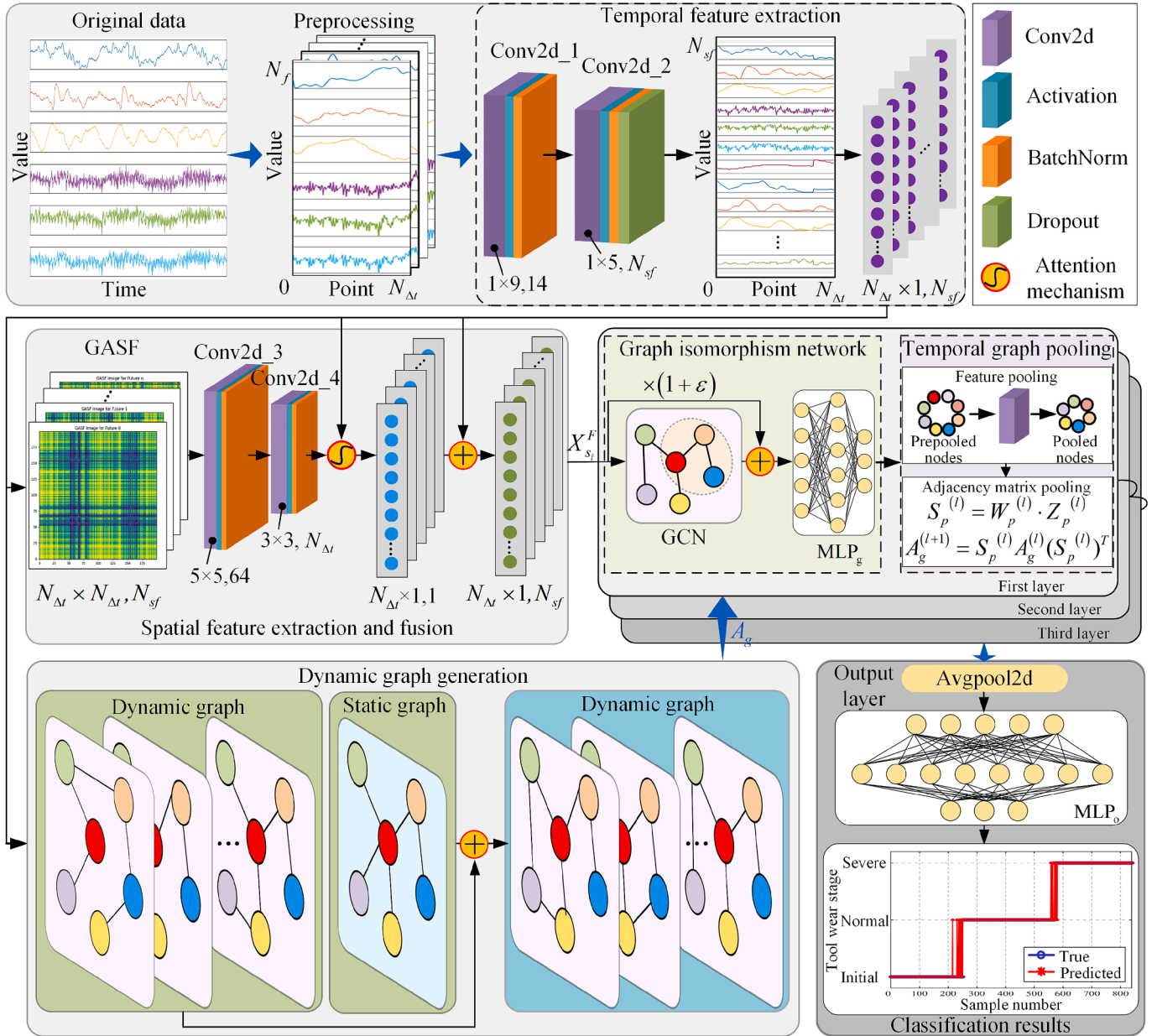


Fig. 1. Overall framework of the tool wear monitoring model.

the conclusions are drawn in Section 5.

2. Monitoring methodology

2.1. Overall framework of the monitoring model

The paper proposes a novel method for categorizing and monitoring the tool wear stages. The monitoring signals during machining process are utilized as input to identify the three distinct types of tool wear states. The general framework of the monitoring model is shown in Fig. 1. Among them, the time series feature extraction module mines the depth dependence between milling force and vibration signals under a potential high-dimension space. The spatial feature extraction module transforms the time series data into a two-dimensional image by the GASF, and further utilizes CNN to extract the embedded spatial features of tool wear. Based on the importance of each temporal feature, relevant features are extracted from spatial features through an attention mechanism and supplemented into the temporal features. The dynamic graph generation module takes the overall invariance and local time-

varying nature of the images into account, and constructs the static and dynamic graphs to establish the spatial correlation of different features. Finally, the GIN module is used to aggregate the features of current node and its neighboring nodes to obtain the tool wear-dependent features, and to realize the classification monitoring of tool wear stage through the TGP.

2.2. Temporal-spatial feature extraction

2.2.1. Temporal feature extraction

The multivariate time series data collected during tool wear monitoring is defined as $D_t = \{X_1, X_2, \dots, X_{s_i}, \dots, X_{N_s}\} \in \mathbb{R}^{N_s \times N_f \times N_{\Delta t}}$, where N_s represents the number of samples, N_f denotes the feature dimension of samples, and $N_{\Delta t}$ indicates the time series length of each feature. Based on the flank wear value VB of cutting tool, the tool wear stage can be classified into three stages: initial wear, normal wear, and severe wear. These classifications are also used to assign corresponding labels $L_t = \{y_1, y_2, \dots, y_{s_i}, \dots, y_{N_s}\} \in \mathbb{R}^{N_s \times 1}$ for each sample. Actually, the objective of

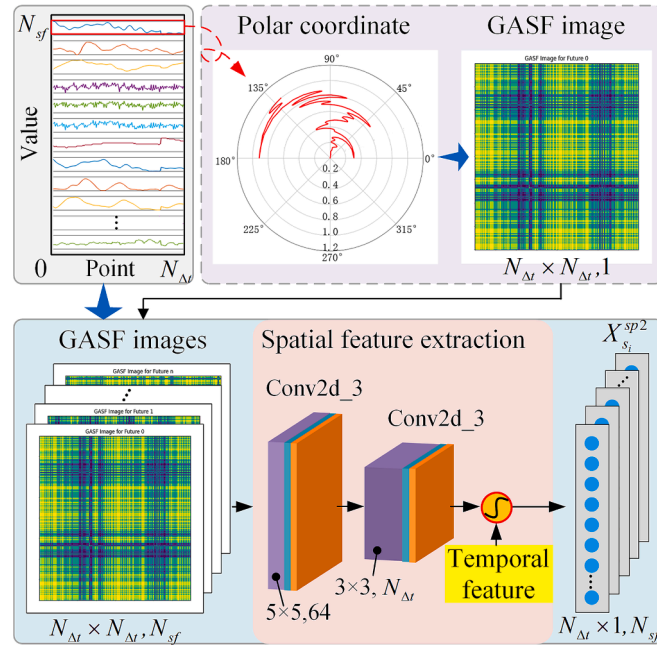


Fig. 2. Spatial feature extraction by GASF and CNN.

tool wear stage classifying is to train a classifier that can recognize the label L_t by the multivariate time series data D_t .

For a single sample $X_{s_i} = \{x_1, x_2, \dots, x_{f_i}, \dots, x_{N_f}\} \in \mathbb{R}^{N_f \times N_{\Delta t}}$ in the preprocessed raw data D_t , in which $x_{f_i} = \{x_{f_i,1}, x_{f_i,2}, \dots, x_{f_i,t_k}, \dots, x_{f_i,N_{\Delta t}}\} \in \mathbb{R}^{1 \times N_{\Delta t}}$ represents a single feature. Considering that a single feature is not yet clear enough in the time dimension, and there are too few nodes for the feature, it is not convenient for the subsequent graph generation module to construct rich spatial relationships. Therefore, through the following equation, the temporal feature extraction is performed to fully exploit the dependencies with its dimension expanded from N_f to N_{sf} . Finally, the improved sequence features $D_{sf} = \{X_1^{sf3}, X_2^{sf3}, \dots, X_{s_i}^{sf3}, \dots, X_{N_s}^{sf3}\} \in \mathbb{R}^{N_s \times N_{sf} \times N_{\Delta t}}$ is obtained.

$$\begin{cases} X_{s_i}^{sf0} = \text{DimTran}(X_{s_i}, \text{dim1}, \text{dim2}) \\ X_{s_i}^{sf1} = \text{Conv2d_1}(X_{s_i}^{sf0}) \\ X_{s_i}^{sf2} = \text{Conv2d_2}(X_{s_i}^{sf1}) \\ X_{s_i}^{sf3} = \text{DimTran}(X_{s_i}^{sf2}, \text{dim1}, \text{dim2}) \end{cases} \quad (1)$$

where Conv2d represents the convolution operation. When doing specific convolution, the computation for the c th channel is given by $O_c(i, j) = \sum_{k=0}^{C_{in}-1} \sum_{m=0}^{K_{in}-1} \sum_{n=0}^{K_{in}-1} W_{c,k}(m, n) \cdot X_k(i \cdot s + m - p, j \cdot s + n - p) + b_c$, in which $W_c \in \mathbb{R}^{C_{out} \times C_{in} \times K_m \times K_n}$ denotes the convolution kernel, $X \in \mathbb{R}^{C_{in} \times H \times W}$ is the input feature, $O \in \mathbb{R}^{C_{out} \times H \times W}$ represents the output feature, i and j correspond to the timestamps of the output features, s is the stride, p is the padding, and b_c is the bias term. Additionally, the ReLU activation function is applied after the convolution to adding nonlinearity to solve the problem of gradient vanishing. Batch normalization is also employed to enhance the training speed and improve generalization capability.

Furthermore, the dropout regularization is utilized to randomly deactivate neurons during training, thereby preventing overfitting.

2.2.2. Spatial feature extraction

The acquired cutting force and vibration signals are one-dimensional time series data. When employing CNN to learn features within time domain, it is difficult to extract high-dimensional temporal features. To address this limitation, the GASF is utilized to encode the one-dimensional signals into two-dimensional data, thereby enriching the temporal information. In the GASF images, the position and color of pixel blocks encapsulate a high-dimensional and comprehensive mapping of the one-dimensional time series signal, thereby rendering the feature extraction via CNN will be more effective. Firstly, normalizing all the sampling points $x_{f_i}^{sf3} = \{x_{f_i,1}^{sf3}, x_{f_i,2}^{sf3}, \dots, x_{f_i,t_k}^{sf3}, \dots, x_{f_i,N_{\Delta t}}^{sf3}\} \in \mathbb{R}^{1 \times N_{\Delta t}}$ of each feature can prevent input and weight polarization and speed up the convergence of the model. The formula is as follows.

$$\tilde{x}_{f_i}^{sf3} = \frac{(2x_{f_i}^{sf3} - \max(x_{f_i}^{sf3}) - \min(x_{f_i}^{sf3}))}{\max(x_{f_i}^{sf3}) - \min(x_{f_i}^{sf3})} \in [-1, 1] \quad (2)$$

Then, as shown in Fig. 2, the normalized $\tilde{x}_{f_i}^{sf3}$ is mapped on polar coordinates and the formula is calculated as follows.

$$\begin{cases} r_{f_i} = [1/N_{\Delta t}, 2/N_{\Delta t}, \dots, 1] \\ \varphi_{f_i} = \arccos(\tilde{x}_{f_i}^{sf3}) \end{cases} \quad (3)$$

Subsequently, the inner product formula defined by the GASF is employed to reconstruct the polar coordinates, which essentially transforms the data into the cosine function of the sum of polar angles between sampling points at arbitrary two time. The final GASF computation formula can be expressed as:

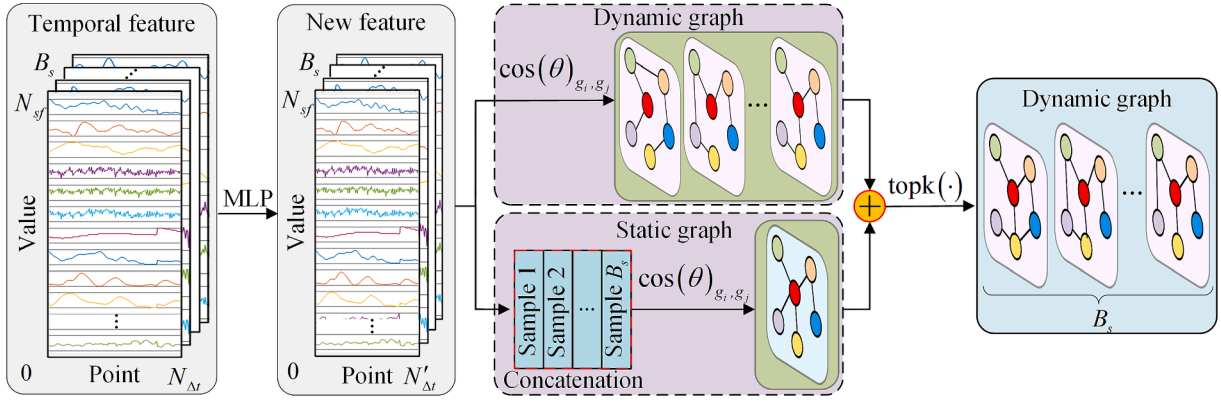


Fig. 3. Generation and fusion process of the static and dynamic graphs.

$$\text{GASF} = \begin{pmatrix} \cos(\varphi_{f_{i,1}} + \varphi_{f_{i,1}}) & \cdots & \cos(\varphi_{f_{i,1}} + \varphi_{f_{i,t_{k2}}}) & \cdots & \cos(\varphi_{f_{i,1}} + \varphi_{f_{i,N_{\Delta t}}}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\varphi_{f_{i,t_{k1}}} + \varphi_{f_{i,1}}) & \cdots & \cos(\varphi_{f_{i,t_{k1}}} + \varphi_{f_{i,t_{k2}}}) & \cdots & \cos(\varphi_{f_{i,t_{k1}}} + \varphi_{f_{i,N_{\Delta t}}}) \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\varphi_{f_{i,N_{\Delta t}}} + \varphi_{f_{i,1}}) & \cdots & \cos(\varphi_{f_{i,N_{\Delta t}}} + \varphi_{f_{i,t_{k2}}}) & \cdots & \cos(\varphi_{f_{i,N_{\Delta t}}} + \varphi_{f_{i,N_{\Delta t}}}) \end{pmatrix} \in \mathbb{R}^{N_{\Delta t} \times N_{\Delta t}} \quad (4)$$

In the study, the matrix size of GASF data is $N_{\Delta t} \times N_{\Delta t}$. Furthermore, since each feature can generate one image, and a single sample comprises N_{sf} features, a total of N_{sf} images can be produced. Next, the CNN is used to extract features from the GASF matrix. As shown in Eq. (6), this process involves two convolutions. To fuse temporal and spatial features, this paper employs a cross-attention mechanism. Attention scores α_{s_i} are computed by calculating the dot product between temporal and spatial features, which are then used to weight and filter the spatial features. The spatial features $X_{s_i}^{sp2} \in \mathbb{R}^{N_{sf} \times N_{\Delta t}}$ were ultimately extracted.

$$\begin{cases} X_{s_i}^{sp0} = \text{Conv2d}_3(\text{GASF}) \\ X_{s_i}^{sp1} = \text{Conv2d}_4(X_{s_i}^{sp0}) \\ \alpha_{s_i} = X_{s_i}^{sf3} * \left(\text{Reshape}(X_{s_i}^{sp1}, (N_{\Delta t}, H \times W)) \right) \\ X_{s_i}^{sp2} = \alpha_{s_i} * \left(\text{Reshape}(X_{s_i}^{sp1}, (N_{\Delta t}, H \times W)) \right)^T \end{cases} \quad (5)$$

Finally, the filtered spatial features $X_{s_i}^{sp2}$ are combined with the temporal features $X_{s_i}^{sf3}$ to achieve a dynamic joint representation of both. $X_{s_i}^F$ is calculated as follows.

$$X_{s_i}^F = X_{s_i}^{sf3} + X_{s_i}^{sp2} \in \mathbb{R}^{N_{sf} \times N_{\Delta t}} \quad (6)$$

2.3. Static and dynamic graphs generation and fusion

In conventional graph theory, the graphs usually have fixed connection relationships. However, when monitoring signals acquired from tool wear experiments, predefined graph structures are absent. Hence, one needs to construct a graph from scratch. In this study, the extracted feature variables are designated as nodes with their corresponding feature values are assigned as node attributes, thereby establishing the spatial correlations. The time-series data, such as cutting force and vibration signals, exhibit a combination of stability and variability by nature. From a holistic perspective, the connection relationships among feature variables remain consistent, representing the static

information. However, as time progresses, the node attributes continuously evolve, leading to subtle variations in the connection relationships at each time interval. This constitutes the dynamic information. Consequently, the study considers the stable and variable characteristics of series feature simultaneously, constructing the static and dynamic graph matrices. Lastly, fusing these two matrices yields an optimal dynamic graph, which can reveal the significant and latent spatial correlations among these feature variables.

2.3.1. Static graph construction

As shown in Fig. 3, the time series feature data is first linear transformed by a multilayer perceptron (MLP), which recombines and compresses the input data into a new representation space $X_g \in \mathbb{R}^{N_{sf} \times N_{\Delta t}}$.

$$X_g = W_g X_{s_i}^{sf3} + b_g \quad (7)$$

where $W_g \in \mathbb{R}^{N_{\Delta t} \times N_{\Delta t}}$ is the weight matrix. $b_g \in \mathbb{R}^{N_{\Delta t} \times 1}$ is the bias vector.

During the static graph construction, all samples $X_g = \{x_{g_1}, \dots, x_{g_i}, \dots, x_{g_j}, \dots, x_{g_{N_{sf}}}\}$ within the batch B_s are concatenated along the temporal dimension, resulting in a concatenated sequence of length $B_s \times N_{\Delta t}$. Subsequently, the cosine similarity between arbitrary two features is computed using Eq. (8). A value closer to one indicates a higher similarity between the features, thereby increasing the likelihood of them being designated as the neighboring nodes.

$$\cos(\theta)_{g_i, g_j} = \frac{\sum_{t_k=1}^{B_s \times N_{\Delta t}} (x_{g_i, t_k} \times x_{g_j, t_k})}{\sqrt{\sum_{t_k=1}^{B_s \times N_{\Delta t}} (x_{g_i, t_k})^2} \times \sqrt{\sum_{t_k=1}^{B_s \times N_{\Delta t}} (x_{g_j, t_k})^2}}, \quad 1 \leq i, j \leq N_{sf} \quad (8)$$

Finally, the adjacency matrix of static graph can be obtained as follows.

$$A_{sg}^{B_s} = \begin{bmatrix} \cos(\theta)_{g_1, g_1} & \cdots & \cos(\theta)_{g_1, g_j} & \cdots & \cos(\theta)_{g_1, N_{sf}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\theta)_{g_i, g_1} & \cdots & \cos(\theta)_{g_i, g_j} & \cdots & \cos(\theta)_{g_i, N_{sf}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\theta)_{N_{sf}, g_1} & \cdots & \cos(\theta)_{N_{sf}, g_j} & \cdots & \cos(\theta)_{N_{sf}, N_{sf}} \end{bmatrix} \quad (9)$$

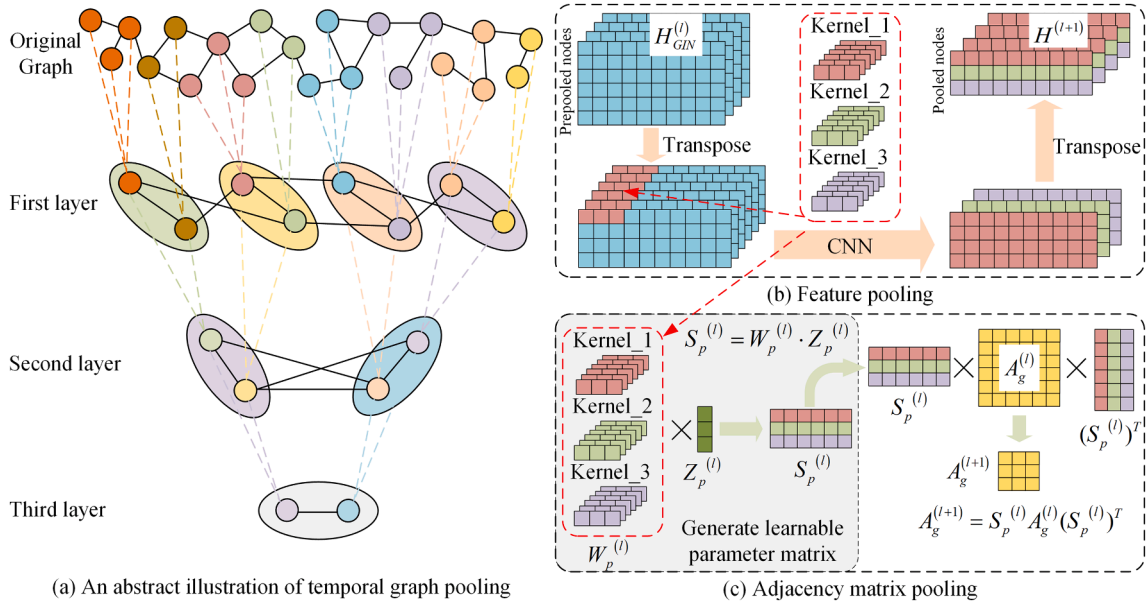


Fig. 4. Schematic diagram of feature matrix pooling and adjacency matrix pooling.

2.3.2. Dynamic graph construction

In the construction of dynamic graph, for the samples X_g within batch B_s , the concatenation is not required. Instead, the cosine similarity between arbitrary two features is calculated directly, so that each sample has a similarity matrix with the following formula.

$$\cos(\theta)_{g_i, g_j} = \frac{\sum_{t_k=1}^{N_{\Delta t}} (x_{g_i, t_k} \times x_{g_j, t_k})}{\sqrt{\sum_{t_k=1}^{N_{\Delta t}} (x_{g_i, t_k})^2} \times \sqrt{\sum_{t_k=1}^{N_{\Delta t}} (x_{g_j, t_k})^2}}, \quad 1 \leq i, j \leq N_{sf} \quad (10)$$

Further, the adjacency matrix of dynamic graph is achieved as $A_{dg}^{B_s} \in \mathbb{R}^{B_s \times N_{sf} \times N_{sf}}$, where the b_s th matrix in the batch size can be computed as follows.

$$A_{dg}^{B_s}(b_s, :, :) = \begin{bmatrix} \cos(\theta)_{g_1, g_1} & \cdots & \cos(\theta)_{g_1, g_j} & \cdots & \cos(\theta)_{g_1, N_{sf}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\theta)_{g_i, g_1} & \cdots & \cos(\theta)_{g_i, g_j} & \cdots & \cos(\theta)_{g_i, N_{sf}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \cos(\theta)_{N_{sf}, g_1} & \cdots & \cos(\theta)_{N_{sf}, g_j} & \cdots & \cos(\theta)_{N_{sf}, N_{sf}} \end{bmatrix} \quad (11)$$

Finally, within a single batch, the adjacency matrix of the static graph and the adjacency matrices of the dynamic graphs for each sample are summed to obtain the combined adjacency matrix $A_g^{B_s} = A_{sg}^{B_s} + A_{dg}^{B_s}$. In order to reduce the computation costs, the top k largest values in the matrix are identified and set to ones, while the remaining elements are set to zeros, resulting in a sparse adjacency matrix $A_g^{B_s} \in \mathbb{R}^{B_s \times N_{sf} \times N_{sf}}$. The calculation steps are shown in the following equation. The above process is applied to multiple batches across the entire dataset, ultimately yielding the adjacency matrix $A_g \in \mathbb{R}^{N_s \times N_{sf} \times N_{sf}}$ for each sample.

$$\begin{cases} \text{index} = \text{topk}(A_g^{B_s}[:, :]) \\ A_g^{B_s}(\text{index}) = 1 \\ A_g^{B_s}(\sim \text{index}) = 0 \end{cases} \quad (12)$$

where $\text{topk}(\cdot)$ is the arithmetic function used to find the first k maximum values.

2.4. Time graph pooling based on GIN

2.4.1. GIN model

Compared with GCN and GAT, the GIN can exhibit higher sensitivity in capturing graph structural features. Therefore, the paper leverages GIN for more in-depth feature extraction. The inputs of GIN are the feature matrix $X_{s_i}^F$ and the adjacency matrix A_{g, s_i} . The node update process consists of two steps. The first step involves aggregating the features of a node and its neighboring nodes, and the second one involves fitting a single mapping function using a MLP. The update formula is as follows.

$$H_{GIN}^{(l)} = \text{MLP}_g^{(l)} \left((1 + \epsilon^{(l)}) \cdot H^{(l-1)} + \tilde{A}_g^{(l)} H^{(l-1)} \right) \quad (13)$$

where $H_{GIN}^{(l)}$ denotes the hidden features of the l th layer after processing by GIN. $H^{(l-1)}$ represents the hidden features of the $(l-1)$ th layer, in which $H^{(0)} = X_{s_i}^F$. $\tilde{A}_g^{(l)} = D_g^{-\frac{1}{2}} A_g^{(l)} D_g^{-\frac{1}{2}}$ is the normalized adjacency matrix of the l th layer, in which $A_g^{(0)} = A_{g, s_i}$, and D_g is the degree matrix of A_g . ϵ is the weight parameter that determines the relative importance of a node's own features compared to its neighboring nodes.

2.4.2. TGP model

For the graph classification tasks, the graph pooling operation can be employed to reduce the number of nodes, extract high-dimensional features, and enhance the classification accuracy. However, some conventional pooling methods, such as max pooling, mean pooling, and sum pooling, would aggregate feature vectors or matrices into a few features. The resulting graph representation vectors tend to be overly smooth, neglecting the potential hierarchical structures within the graph and failing to learn the hierarchical representations of graph. In this regard, the study adopt the TGP method. The method utilizes the temporal convolutions to cluster nodes and extract temporal features, followed by reconstructing the adjacency matrix using convolutional weights. By integrating graph pooling with temporal processing, it not only reduces the feature dimensionality but also preserves the key temporal characteristics. This makes it particularly suitable for handling the graph data with temporal dependencies.

As depicted in Fig. 4(a), the network architecture employs a three-layer TGP scheme. In each layer, the output $H_{GIN}^{(l)}$ of the GIN and the adjacency matrix $A_g^{(l)}$ are separately subjected to pooling operations, with the objective of progressively reducing the number of nodes in a layer-wise manner. Fig. 4(b) illustrates the pooling process of the feature matrix. Herein, the number of remaining nodes is controlled by setting the pooling rate autonomously, and the pooling of the feature matrix is implemented using a CNN. The computation of the pooled hidden features $H^{(l+1)}$ is given by the following equation. During the convolution operation, the kernel parameter is $W^{(l)} \in \mathbb{R}^{N_{pos}^{(l+1)} \times N_{pre}^{(l)} \times 1 \times S_{kn}^{(l)}}$, where $N_{pre}^{(l)}$ denotes the number of nodes before pooling, $N_{pos}^{(l+1)}$ represents the number of nodes after pooling, and $S_{kn}^{(l)}$ is the convolutional kernel width of layer l .

$$\begin{cases} H_{GIN}^{(l)} = \text{DimTran}(H_{GIN}^{(l)}, \text{dim1}, \text{dim2}) \\ H^{(l+1)} = \text{Conv2d}(H_{GIN}^{(l)}) \\ H^{(l+1)} = \text{DimTran}(H^{(l+1)}, \text{dim1}, \text{dim2}) \end{cases} \quad (14)$$

Fig. 4(c) shows the pooling process of the adjacency matrix. Firstly, the learnable parameter vector $Z_p^{(l)} \in \mathbb{R}^{S_{kn}^{(l)} \times 1}$ is randomly generated to obtain the matrix $S_p^{(l)}$, as follows.

$$S_p^{(l)} = W_p^{(l)} \cdot Z_p^{(l)} \quad (15)$$

In turn, one calculates the pooled adjacency matrix $A_g^{(l+1)} \in \mathbb{R}^{N_{pos}^{(l+1)} \times N_{pre}^{(l)}}$, as shown in the following equation.

$$A_g^{(l+1)} = S_p^{(l)} A_g^{(l)} (S_p^{(l)})^T \quad (16)$$

Additionally, during the pooling process, the batch normalization is applied to the feature matrix output from pooling, thereby enhancing both training speed and generalization capability. The ReLU activation function is employed to introduce nonlinearity, which can effectively address the vanishing gradient problem. To prevent overfitting during training, some neurons are randomly dropped through regularization. In the subsequent layer iteration, $H^{(l+1)}$ and $A_g^{(l+1)}$ are substituted into Eqs. (13)–(16), and through the multiple layer iteration, the computational capacity can be significantly improved. After undergoing three layers of GIN and TGP processing, each node would aggregate the dominated features of its nearby information, thereby providing a more comprehensive representation of the graph properties.

2.5. Output of the monitoring model

Based on the beforementioned pooled outputs, an adaptive average pooling layer and a linear layer are employed to generate the predicted probabilities $y_p \in \mathbb{R}^{N_s \times 3}$ of tool wear condition categories for all samples. To obtain the predicted classification results of tool wear stage, the index corresponding to the maximum value in y_p is identified, which represents the predicted outcome $L_p \in \mathbb{R}^{N_s \times 1}$. The specific calculation formula is as follows.

$$\begin{cases} y_p = \text{MLP}_o(\text{AdaptiveAvgPool2d}(H^{(3)})) \\ L_p = \text{Argmax}(y_p) \end{cases} \quad (17)$$

where MLP_o is the MLP of the output layer. Argmax is the index function for the maximum value.

The following algorithm 1 shows the computation framework of the entire monitoring model.

Algorithm 1 Proposed algorithm framework.

Input: The tool wear dataset, $D_t = \{X_1, X_2, \dots, X_{st}, \dots, X_{N_s}\} \in \mathbb{R}^{N_s \times N_f \times N_{st}}$

Output: Predict label, $L_p = \{y_1, y_2, \dots, y_{st}, \dots, y_{N_s}\} \in \mathbb{R}^{N_s \times 1}$

```

begin
  for  $X_{st}$  in  $D_t$ ,  $X_{st} \in \mathbb{R}^{N_f \times N_{st}}$  do
     $X_{st}^{sf3} = F(X_{st}) \in \mathbb{R}^{N_{sf} \times N_{st}} \leftarrow \text{Eq.(1)}$  # Temporal feature extraction
     $X_{st}^{sp2} = F(X_{st}^{sf3}) \in \mathbb{R}^{N_{sp} \times N_{st}} \leftarrow \text{Eq.(2-5)}$  # Spatial feature extraction
     $X_{st}^F = X_{st}^{sf3} + X_{st}^{sp2} \in \mathbb{R}^{N_{sf} \times N_{st}} \leftarrow \text{Eq.(6)}$  # Spatiotemporal fusion
     $A_{sg}^{B_s} = F(\text{concatenate}(\text{MLP}(X_1^{sf3}, \dots, X_{st}^{sf3}, \dots, X_{N_s}^{sf3}))) \in \mathbb{R}^{N_{sg} \times N_{sf}}$ 
     $\leftarrow \text{Eq.(7-9)}$  # Static graph
     $A_{dg}^{B_s} = F(\text{MLP}(X_1^{sf3}, \dots, X_{st}^{sf3}, \dots, X_{N_s}^{sf3})) \in \mathbb{R}^{N_{dg} \times N_{sf}}$ 
     $\leftarrow \text{Eq.(7,10,11)}$  # Dynamic graph
     $A_g \leftarrow A_{sg}^{B_s} = A_{sg}^{B_s} + A_{dg}^{B_s} \in \mathbb{R}^{N_{sg} \times N_{sf}}$  # Dynamic graph generation
    for  $l$  in layers, do
       $H_{GIN}^{(l)} = \text{GIN}(A_{g_{st}}, X_{st}^F) \leftarrow \text{Eq.(13)}$  # GIN
       $H^{(l+1)}, A_g^{(l+1)} = \text{TGP}(H_{GIN}^{(l)}, A_g^{(l)}) \leftarrow \text{Eq.(14-16)}$  # TGP
    end
     $y_p = \text{MLP}_o(\text{AdaptiveAvgPool2d}(H^{(3)}))$ 
     $L_p = \text{Argmax}(y_p)$  # Predict label
  end
end

```

The structural hyper-parameters of the proposed network model are shown in Table 1.

3. Experiment

3.1. PHM2010 dataset

The study validates the tool wear stage classification model using the publicly available PHM2010 dataset. As shown in Fig. 5, the experiments were conducted on a Roders Tech RFM 760 high-speed CNC machine. Six three-tooth ball-end tungsten carbide cutters are used to cutting a rectangular workpiece row by row under dry milling conditions. The workpiece material is Inconel 718, and the distance of each cutting path is 108 mm. The specific milling parameters were as follows: a spindle speed of 10400 rpm, a feed rate of 1555 mm/min, a cutting depth of 0.2 mm, a cutting width of 0.125 mm. During the cutting process, a Kistler three-channel gauge acquired the milling force signal, a three-channel piezoelectric accelerometer gathered the workpiece vibration signal, and an acoustic emission sensor collected the AE signal. The raw data contained seven channels. The multiple signals were amplified and processed using a Kistler 5019A charge amplifier. Then, these data were synchronized by a NI DAQ card with a sampling frequency of 50 kHz. After each cutting pass, the wear value on the tool flank face is measured using a LEICA MZ12 microscope. Under these cutting conditions, the full-life tool wear experiments were repeated for the six tools numbered C1–C6. The experimental data were collected by 315 tool walks for each tool.

Given that only the datasets for tools C1, C4, and C6 contain measurement values of flank wear for milling cutters among the dataset, the study exclusively utilizes these three datasets for both analysis and validation. Fig. 6(a) illustrates the trend of flank wear for the C1 tool. It is evident that the wear width varies among the three cutting edges and exhibits a nonlinear pattern. Consequently, assessing the overall wear condition of the tool based solely on the wear value of a single tooth is inaccurate. Therefore, the study adopts the average wear value of the

Table 1
Hyper-parameters of the proposed neural network model.

Model	Layer	Details	Input data size	Output data size
Temporal feature extraction	Conv2d_1	Transpose(1,2), $F_s = 14$, $K_s = (1,9)$, $P_d = (0,4)$, ReLU, BatchNorm2d	[4,1,6,288]	[4,14,1,288]
	Conv2d_2	$F_s = 24$, $K_s = (1,5)$, $P_d = (0,2)$, ReLU, BatchNorm2d, Dropout = 0.5, Transpose(1,2),	[4,14,1,288]	[4,1,24,288]
Spatial feature extraction and fusion	GASF	Image size = 288	[4,1,24,288]	[4,24,288,288]
	Conv2d_3	$F_s = 64$, $K_s = 5$, ReLU, BatchNorm2d	[4,24,288,288]	[4,64,285,285]
	Conv2d_4	$F_s = 288$, $K_s = 3$, ReLU, BatchNorm2d, Dropout = 0.5	[4,64,285,285]	[4,288,284,284]
	Attention score	[4,1,24,288] → [4,24,288] [4,288,284,284] → [4,288,80656]	[4, 24,288] [4,288,80656]	[4,24,80656]
	Spatial feature Fusion	[4,288,80656] ^T [4,24,288] → [4,1,24,288]	[4,24,80656] [4,80656,288] [4,1,24,288] [4,1,24,288]	[4,24,288] [4,1,24,288]
Static and dynamic graph generation and fusion	MLP	$H_{id} = 256$, $H_{out} = 64$, ReLU	[4,1,24,288]	[4,1,24,64]
	Static	Concatenate in time dimension	[4,1,24,64]	[1,1,24,24]
	Dynamic	Without concatenate	[4,1,24,64]	[4,1,24,24]
	Fusion	Broadcasting, Topk = 288	[1,1,24,24] [4,1,24,24]	[4,1,24,24]
GIN and TGP	First layer	Eps = 0.5, $F_s = 32$, $K_s = 1$, bias = True	[4,1,24,24] [4,1,24,288]	[4,32,24,288]
		$F_s = 19$, $K_s = (1,18)$, $S_d = 1$, $P_d = (0,8)$	[4,1,24,24] [4,32,24,288]	[4,1,19,19] [4,32,19,288]
	Second layer	Eps = 0.5, $F_s = 64$, $K_s = 1$, bias = True	[4,1,19,19] [4,32,19,288]	[4,64,19,288]
		$F_s = 14$, $K_s = (1, 9)$, $S_d = 1$, $P_d = (0, 4)$	[4,1,19,19] [4,64,19,288]	[4,1,14,14] [4,64,14,288]
	Third layer	Eps = 0.5, $F_s = 128$, $K_s = 1$, bias = True	[4,1,14,14] [4,64,14,288]	[4,128,14,288]
		$F_s = 10$, $K_s = (1, 5)$, $S_d = 1$, $P_d = (0, 2)$	[4,1,14,14] [4,128,14,288]	[4,1,10,10] [4,128,10,288]
Output layer	Adaptive AvgPool2d	Output size = 1	[4,128,10,288]	[4,128,1,1]
	MLP	$H_{in} = 128$, $H_{out} = 3$	[4,128]	[4,3]
	Argmax	Dim = 1	[4,3]	[4,1]

three cutting edges to delineate the wear stages [46]. Fig. 6(b) presents the full-life wear curves for the three tools. The wear curves across the three datasets exhibit differences in both numerical values and local

trends. As shown in Fig. 6(c), which displays the wear values at each time point, all three tools entered the normal wear phase around the 50th pass. After the 210th pass, the wear curves began to fluctuate more

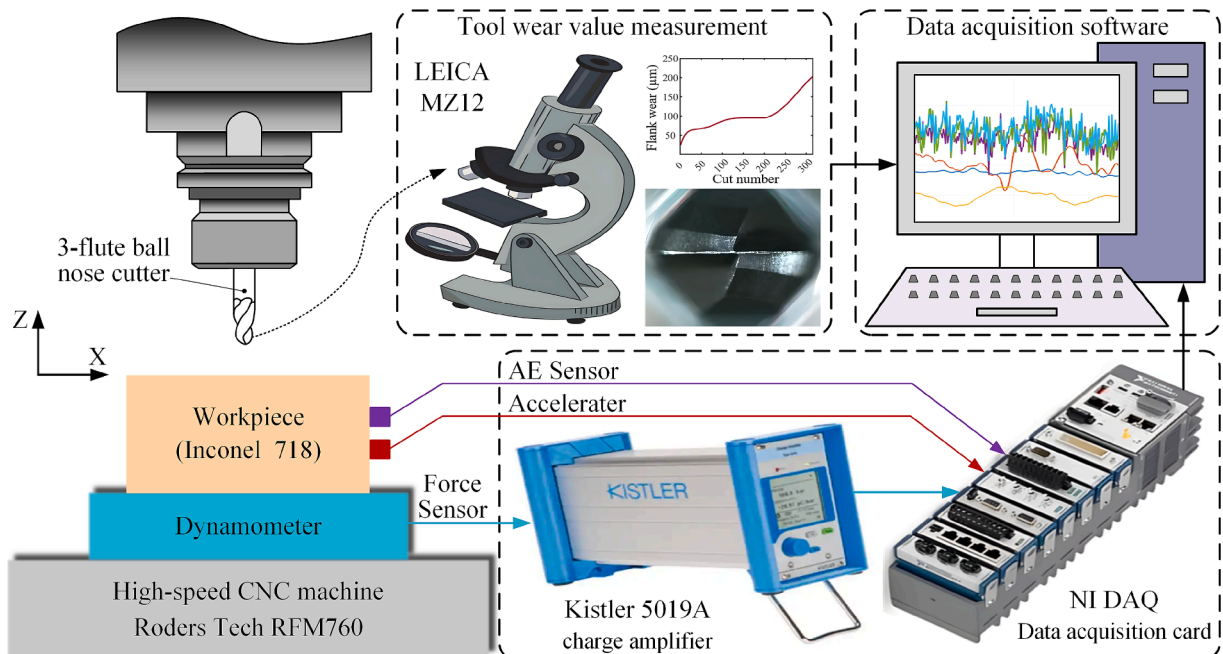


Fig. 5. Schematic diagram of the experimental procedure for the PHM2010 dataset.

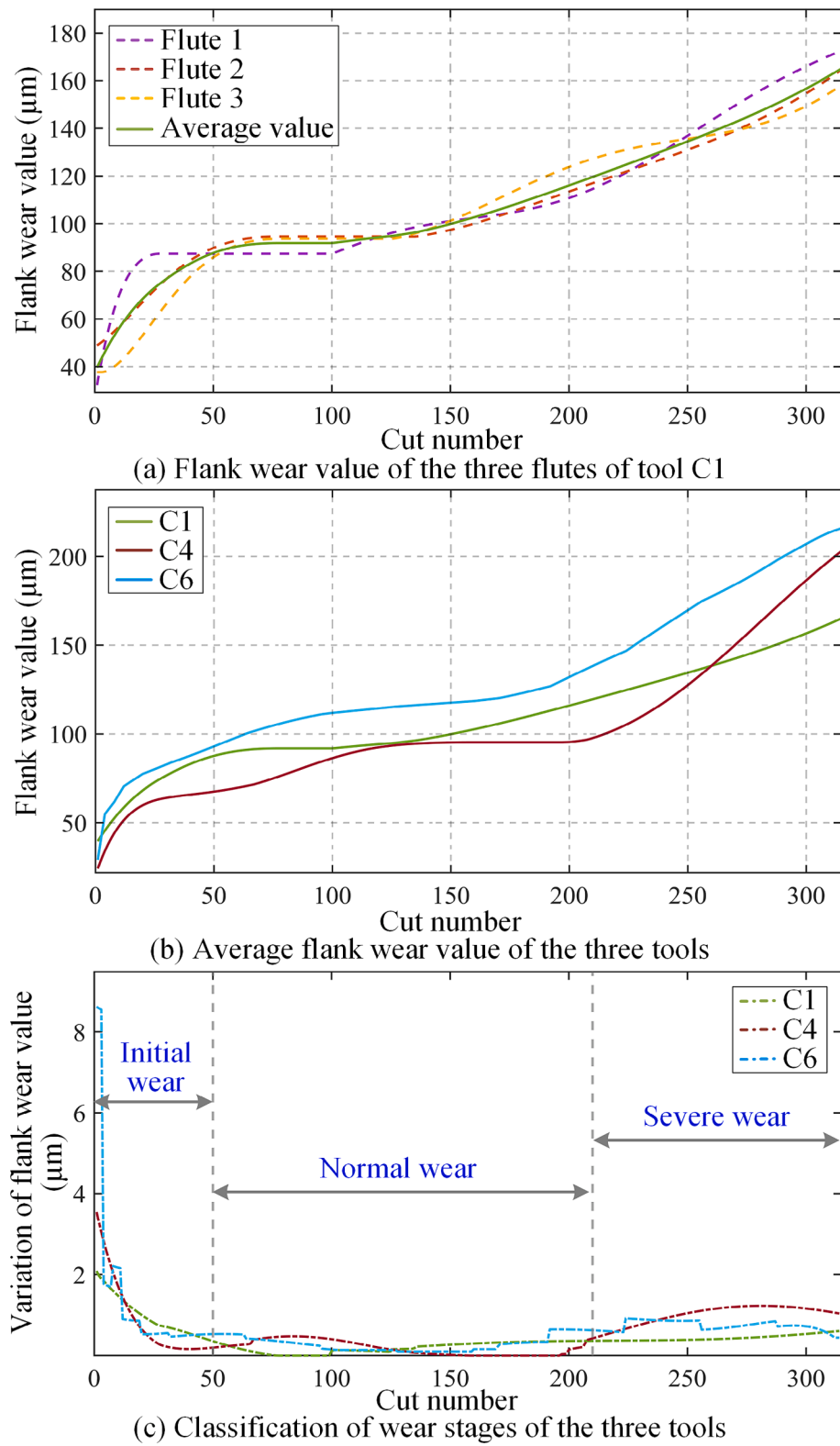


Fig. 6. The measured data of tool wear values.

significantly, with some data points exhibiting abnormal variations, indicating the onset of the severe wear phase. Therefore, the study categorizes the wear stages as follows: the (1–50)th passes are defined as the initial wear stage, the (51–210)th passes are defined as the normal wear stage, and the (211–315)th passes are defined as the severe wear stage.

3.2. Dataset preprocessing

Considering the extremely severe wear and unstable signals in the later stages of each milling, the first 300 walking tool data of each data set are selected in this paper, and discard the sound emission signals with more noise. Besides, during the in-cutting and out-cutting processes

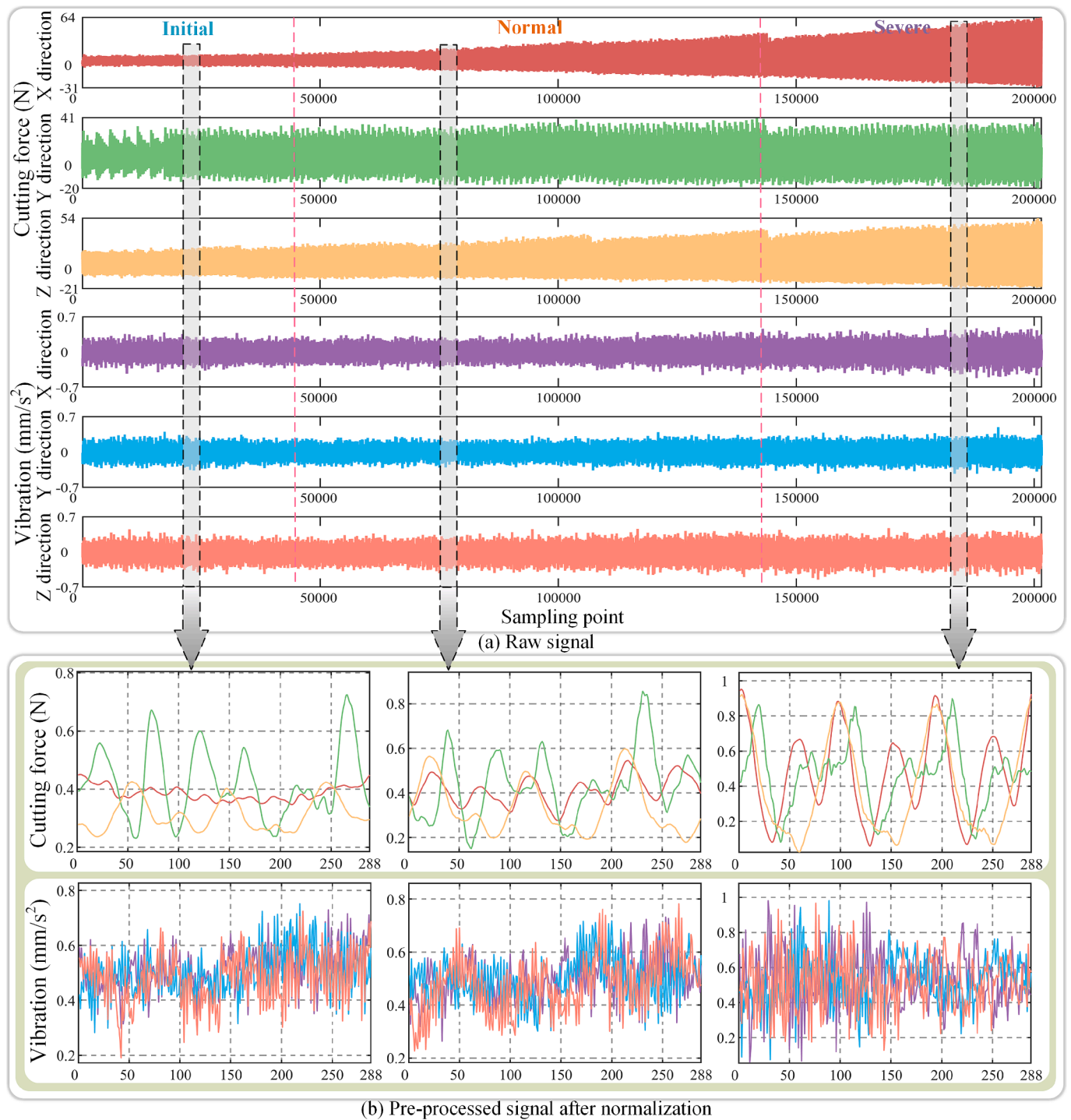


Fig. 7. Data preprocessing of cutting force and vibration signal.

of each tool walking, the contact between the tool teeth and the workpiece is incomplete, and the signals collected by the sensors will fluctuate significantly, thus affecting the training and identification of monitoring model. In order to reduce the interference of unsteady segment, the paper removes the initial 40,000 and final 40,000 data points during each tool cutting feed, and only retains the data within the stable cutting. Divide the stable phase data into 10 equal portions. Based on variations in pass counts across wear stages (e.g., the stable stage having the highest pass count), multiple samples of length 288 (one rotational cycle) were extracted from each portion. For example, if the initial wear stage has 50 cutting passes, collect 288-length sample points

from each of the first five segments, ultimately yielding 250 samples. The raw data of the vibration and force signal and its preprocessing process are shown in Fig. 7, from which several equal-length samples are taken after the stabilized cutting portion is obtained from the raw data.

In addition, the tool wear curves show that the normal wear stage data is significantly higher than that of the initial and severe wear stages, resulting in an obvious data imbalance of the three stages. To alleviate this problem, the study adopts a stratified sampling strategy with more sampling for the stage with less data and less sampling for the stage with more data, to ensure that the number of samples in each stage tends to be balanced. Specifically, five, two, and three samples were

Table 2
Division of the datasets.

Datasets	Data type
A	Force signals
B	Vibration signals
D	Force and vibration signals
E	Force and vibration signals

Table 3
Division of training and test datasets.

Datasets	Training datasets	Test datasets
A1/B1/D1	C1 and C4	C6
A2/B2/D2	C4 and C6	C1
A3/B3/D3	C6 and C1	C4
E1	C1	C4
E2	C4	C6
E3	C6	C1

collected from each tool traveler data in the initial wear, normal, and severe wear stages, respectively. Ultimately, 250, 320, and 270 samples were obtained for the three stages, respectively. Lastly, a total of 840 samples for a single tool are obtained. Based on the experimental data of the three tools, a total of 2520 samples were constructed in the study, providing a more balanced and representative dataset for subsequent analysis.

The cross datasets A, B, D, and E were established to train and test the monitoring model. The data type partitioning of these datasets is illustrated in Table 2, where A dataset exclusively utilizes the cutting force signals, B dataset solely employs the vibration signals, and D and E datasets incorporate both the cutting force and vibration signals. The experimental partitioning of the datasets is detailed in Table 3. For A, B, and D datasets, the data from arbitrary two tools were used as the training and validation sets, with a partitioning ratio of 0.7:0.3, while the data of the remaining tool served as the test set. For E dataset, the data from one tool was divided into training and validation sets with a ratio of 0.7:0.3, and the data of the other two tools were separately used for testing. Compared to training and testing on the same tool, this division can better evaluate the generalization performance of monitoring model.

3.3. Evaluation indicators

In order to evaluate the classification performance of the model, the confusion matrix and the following five evaluation indicators are used, which are accuracy, precision, recall, specificity and F1 score. The calculation formula is as follows.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (19)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (20)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (21)$$

$$\text{F1score} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (22)$$

where TP is the number of samples correctly predicted by the model to be in the positive category. FP is the number of samples incorrectly predicted by the model to be in the positive category. TN is the number of samples correctly predicted by the model to be in the negative category. FN is the number of samples incorrectly predicted by the model to be in the negative category.

Among the above parameters, the confusion matrix is a tool used to evaluate the performance of a classification model. It visually demonstrates the correct classifications and misclassifications among different categories by comparing the predicted results with the actual labels. The accuracy rate reflects the classification performance across all samples, representing the ratio of the number of correct predictions to the total number of samples. This metric indicates the overall prediction ability of the model. The precision rate measures the accuracy in identifying specific stages of wear and tear, focusing on the reliability of the prediction results. Recall assesses the ability to detect the target class, reflecting its coverage of positive class samples. Specificity is an important measure of the ability to identify negative class samples, particularly useful in addressing class imbalance problems. The F1 score serves as a harmonic mean of the precision and recall rates, providing a combined measure of model robustness, with values closer to one indicating a more balanced classification performance.

3.4. Hyper-parameter optimization of the model

In order to configure the optimal hyper-parameters for the network model, an optimization analysis of the hyper-parameters was performed. The analysis was done on a laboratory server. The server was configured with Windows 11 operating system, 60 GB RAM, NVIDIA GeForce RTX 4090 GPU (16,384 CUDA cores), and the CPU was assigned 18 virtual CPU cores, based on AMD EPYC 9754 processor (128 physical cores). The software versions used were PyCharm 2024.3 (Professional Edition), PyTorch 1.12.0 + cu113, Anaconda3 2023.09 and CUDA version 12.7.

As shown in Fig. 8(a), feature association was measured using cosine similarity, and the impact of different top_k values (multiples of 24) on model performance was tested. Results reveal that test accuracy first rises then falls with top_k: too small a value limits neighbor node information extraction, while too large a value introduces noise,

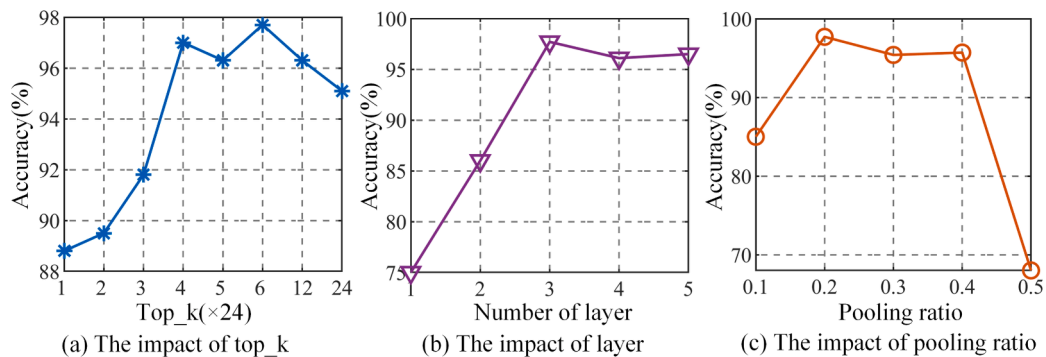


Fig. 8. Analysis and selection of key model parameters.

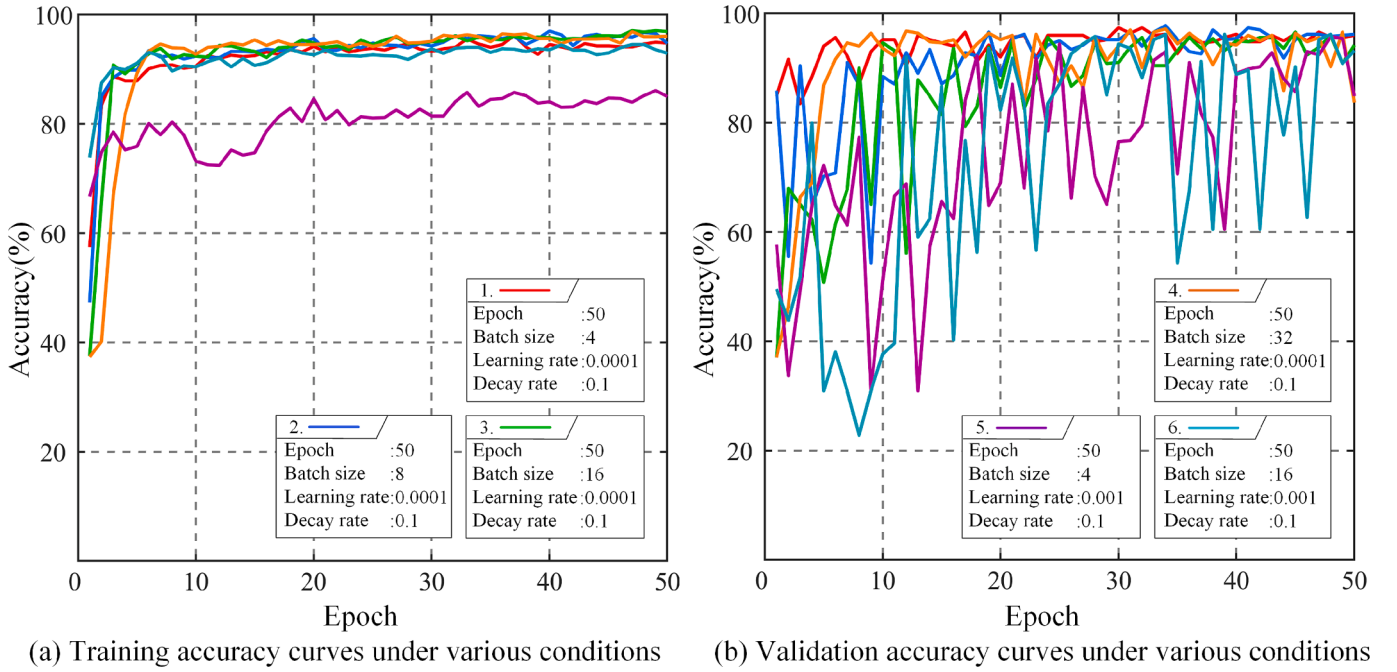


Fig. 9. Accuracy comparison under different hyper-parameter schemes.

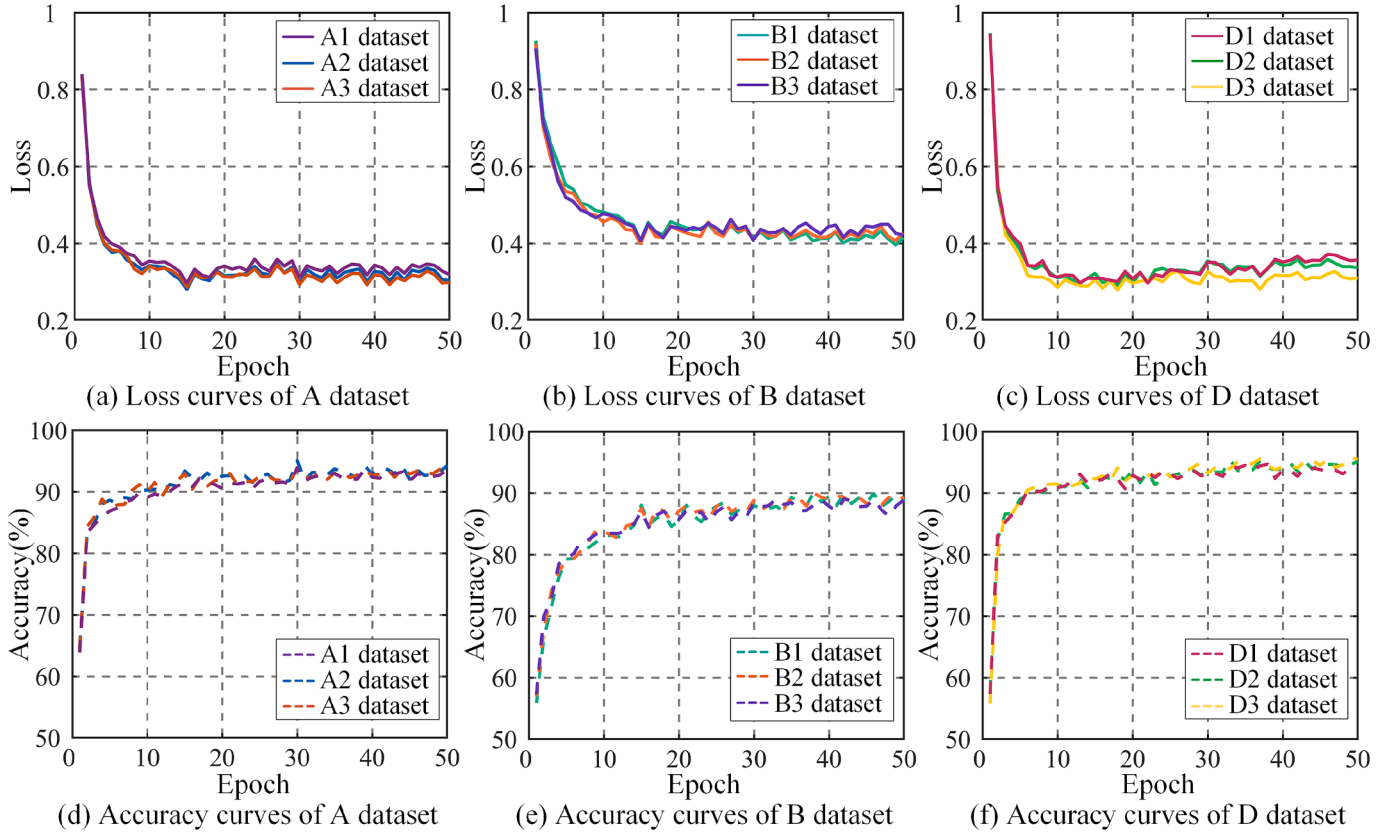


Fig. 10. Loss and accuracy curves for A, B, and D datasets.

degrading performance and increasing computational cost. Thus, top_k was set to 6×24 , balancing efficiency and performance. Fig. 8(b) shows that model accuracy improves and stabilizes as GIN layers increase. Too few layers fail to extract deep features, while too many risk overfitting. A 3-layer GIN module was chosen as optimal. In Fig. 8(c), gradually

increasing the pooling rate shows test accuracy first rising then falling, peaking at 0.2. Moderate pooling extracts key features effectively, while excessive pooling loses important information.

Based on the existing experience of model training, it is known that a larger initial learning rate will lead to difficult reaching the global

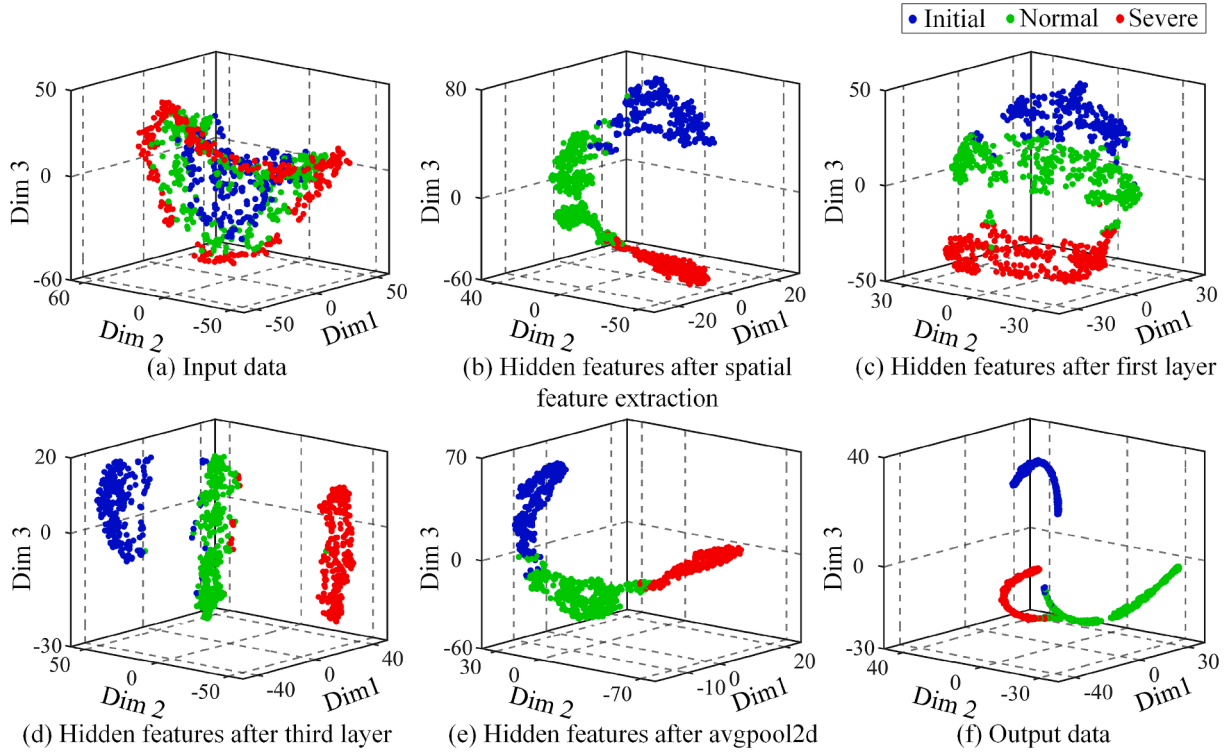


Fig. 11. t-SNE visualization of D3 test set.

optimum and constant oscillation in accuracy during training. If the learning rate is too small, the model will converge more slowly, and require more training time. Additionally, a small batch training generalizes well but leads to some problems, such as low efficiency and difficult convergence. Increasing the batch size can reduce training time, make gradient calculation more stable, and help the model achieve better results during fine-tuning. However, it also increases the risk of overfitting and requires more memory.

Hence, the paper adopts a decay factor to dynamically adjust the learning rate. If there is no improvement in the index after 10 iterations, the learning rate is reduced by half, thus allowing the model to continue to learn the tool wear features. The L2 regularization is used to reduce the overfitting of the model and make the training process smoother, and the regularization factor is set to 0.1. The cross entropy is used as the loss function. The smaller the loss value is, the higher recognition accuracy the model has. The Adam optimizer was used to monitor the iterative backpropagation calculation. Finally, a six hyper-parameter schemes were used for model training, as shown in Fig. 9. The accuracy curves of the training and validation processes are plotted in the figures. The comparison shows that increasing the batch size does not have much effect on the accuracy of the training set, but it leads to large fluctuations in accuracy of the validation set. Besides, increasing the learning rate will reduce the accuracy curves of the training and validation sets at the same time, and the accuracy of validation set will also fluctuate drastically. Combining the above analysis, it is optimal to adopt the hyper-parameter setting of the first scheme.

After determining the above key parameters, the final total number of parameters for the model in this paper is 321,002, which is considered medium-sized. The memory usage for forward/backward propagation is 245.51 MB, accounting for 99.5 % of total memory. The model's total memory requirement is 246.74 MB, making it suitable for running on consumer-grade GPUs (such as NVIDIA GTX series) or high-performance CPUs. This memory requirement is feasible in industrial scenarios. Regarding time complexity, the data packet acquisition time is 0.5 s, the preprocessing time is 0.1129 s, and the model computation time is 0.0296 s. Therefore, the total latency for a single sample is

approximately 0.6425 s, this performance meets the real-time requirements of most industrial scenarios.

4. Validation results

4.1. Convergence process of model training

Based on the optimal parameters obtained in Section 3.4, the model training was performed on A, B, D and E datasets. Among them, the accuracy and loss curves of the training process on A, B and D datasets are shown in Fig. 10. From the figure, the accuracy of the model tends to converge after the 8th iteration, and the loss tends to converge after the 10th iteration. The high accuracy and low loss obtained from training on all four datasets indicate that the model can effectively learn the nonlinear mapping relationship between tool wear values and sensor signals.

The t-SNE algorithm is able to maintain the structure and relevance of data in high-dimensional space. Therefore, the t-SNE is used here to downscale the extracted high-dimensional features in order to visualize their distribution. Fig. 11 shows the 3D visualization of the feature vectors at different stages during the model training process. It can be seen that the input samples are randomly distributed in Fig. 11(a), and there is no obvious correlation between the tool wear stages and the samples. On the contrary, as shown in Fig. 11(b–f), as the model digs deeper into the features of the samples, the strong dependence features of tool wear are gradually extracted, and the differentiation of features between the different wear stages becomes more and more obvious. Especially in the final stage in Fig. 11(f), the samples of three different wear stages are clearly distinguished, which proves the effectiveness of the proposed method.

4.2. Classification results of tool wear stage

After the model training is completed, the tool wear stage recognition model is tested using the test set. Six performance evaluation metrics, including the confusion matrix and accuracy are used to assess

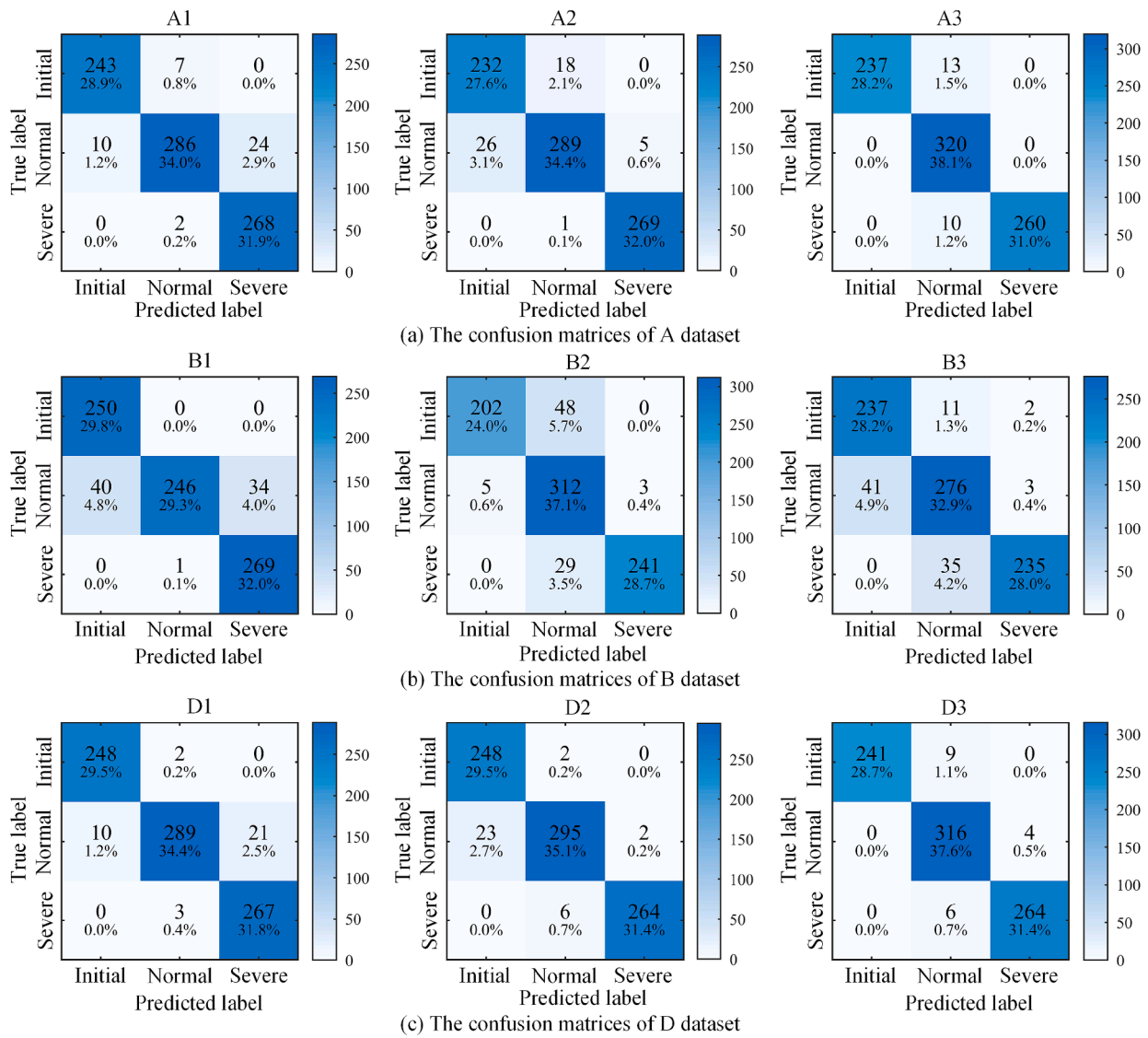


Fig. 12. Confusion matrix of classification results for A, B and D datasets.

the effectiveness of monitoring model. Fig. 12 shows the confusion matrices for A, B, and D datasets. Fig. 13 shows the confusion matrix for E dataset. The confusion matrix enables us to visualize the total number of samples, the distribution of samples across each category, and the counts of correctly and incorrectly classified samples. Each row in the matrix corresponds to the actual category, while each column indicates the predicted category. The integers denote the number of samples, and the percentages illustrate the proportion of these samples relative to the total sample count. The diagonal elements represent the number of samples accurately classified within each category, whereas the off-diagonal elements signify the number of samples that were misclassified and missed by the model.

In tool wear stage identification, minimizing false alarms in the initial and normal wear stages is important to avoid premature tool changes that lead to waste. In the severe wear stage, the focus should be on avoiding missed alarms to prevent late tool changes that will result in lower part machining quality. Thus, the initial and normal wear stages require more attention to the precision, while the severe wear stage requires more attention to the recall and specificity. The evaluation metric values for D, and E datasets are shown in Tables 4 and 5, respectively. The radar charts for A, B, D, and E datasets are shown in Fig. 14. The average precision rates for the four datasets in the initial and normal wear stages were 95.0 %, 89.1 %, 96.5 %, and 94.4 %, respectively.

The combined average recall, specificity, and F1 scores for the severe wear stage were 98.0 %, 94.3 %, 98.0 %, and 98.1 %, respectively. The high combined mean values of recall, specificity, and F1 score in the severe wear stage indicate that the model can effectively distinguish between force signals and non-severe wear. This ability to recognize severe wear in time can prevent production accidents, reduce maintenance costs, and ensure machining safety, which is highly significant in practical applications.

Comparing the results from A, B and D datasets, the average accuracy using only force signal reached 95.4 %, which is 5.4 % higher than the only vibration signal with 90.0 %. This indicates a stronger dependency between force signals and tool wear. Additionally, the presence of more noise in vibration signals is another factor contributing to their lower accuracy. When vibration and force signals were fused, the recognition accuracy increased to 96.5 %, representing improvements of 1.1 % and 6.5 % over the individual signals, respectively. This conclusion demonstrates that the multi-information fusion can significantly outperform the single-signal approaches in terms of recognition accuracy. Notably, the fused approach exhibited more stable performance in accuracy, precision, and recall, suggesting that the force and vibration signals possess complementary characteristics, which significantly enhance the classification performance. Under the E dataset, the average accuracy is 94.7 %, which is slightly lower than that of A and D datasets. This is

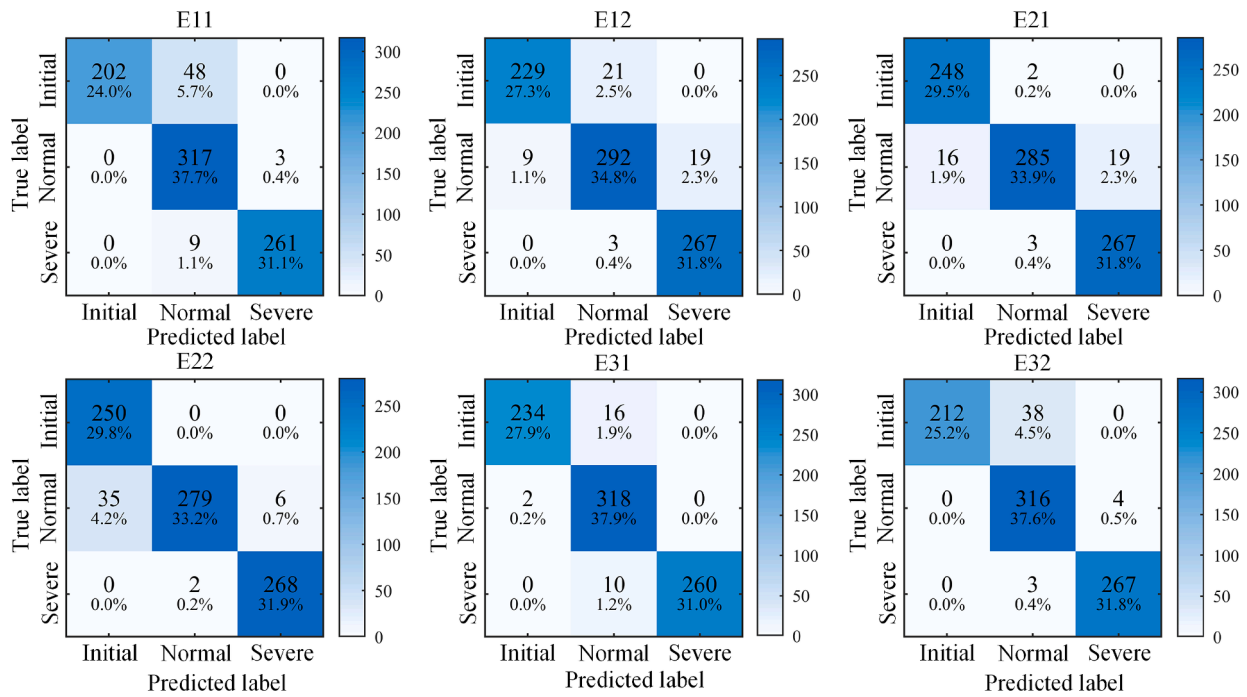


Fig. 13. Confusion matrix of classification results for E dataset.

Table 4

Evaluation indicators on the D dataset.

Indicators		D1	D2	D3
Accuracy (%)		95.7	96.1	97.7
Precision (%)	P _{initial}	96.1	91.5	100.0
	P _{normal}	98.3	97.4	95.5
Recall (%)	R _{severe}	98.9	97.8	97.8
Specificity (%)	S _{severe}	96.3	99.6	99.3
F1 score (%)	F _{severe}	95.7	98.5	98.1

Table 5

Evaluation indicators on the E dataset.

Indicators		E11	E12	E21	E22	E31	E32
Accuracy (%)		92.9	93.8	95.2	94.9	96.7	94.6
Precision (%)	P _{initial}	100.0	96.2	93.9	87.7	99.2	100.0
	P _{normal}	84.8	92.4	98.3	99.3	92.4	88.5
Recall (%)	R _{severe}	96.7	98.9	98.9	99.3	96.3	98.9
Specificity (%)	S _{severe}	99.5	96.7	96.7	98.9	100.0	99.3
F1 score (%)	F _{severe}	97.8	96.0	96.0	98.5	98.1	98.7

attributed to the greater distribution differences between the training and testing sets in E dataset.

Fig. 15 shows the monitoring results of tool wear stages for the A3, B1, D1, and D3 test sets. The left figures represent the probability values of each sample being predicted as three categories, while the right figures show the final classification results of the samples. As can be seen, the overall recognition performance is good, and the misclassified samples are mainly concentrated in the transition region between different wear stages. The primary cause of misclassification lies in the gradual nature of tool wear, which results in no clear distinction between adjacent wear stages. This is corroborated by the fact that the intermediate region can be accurately identified. Furthermore, the tool wear curves reveal that despite identical cutting parameters across all three tools, their wear curves exhibit significant variations due to uncontrollable factors. This poses a significant challenge to accurately delineating the distinct stages of tool wear. In Fig. 15(a–f), the misclassified samples tend to be distributed on one side, likely due to the

inaccurate division of wear stages. Fig. 15(g and h) shows that the model achieves the best classification results in the D3 test set, which has the fewest misidentified samples in the transition region and a more balanced distribution.

In order to fully evaluate the performance of the model, the ROC curve is given. The closer the curve is to the upper left corner, the better the performance of the classifier. In addition, the area under the ROC curve (AUC) provides a comprehensive performance metric for all possible thresholds. Here, the AUC value in a range of [0,1], and the larger AUC values indicates better classifier performance. As shown in Fig. 16, the ROC curves and AUC values are plotted for the D and E datasets, respectively. From the figures, it can be seen that the ROC curves of the proposed model are very close to the upper left corner and the AUC values are over 99.2 %, which indicates that the model has a very well overall performance.

Although the cutting experiments for all three tools were conducted under the same machining conditions, the collected cutting force and vibration signals will differ each time due to factors such as tool mounting errors, changes in workpiece material properties, and environmental disturbances. These differences in the datasets would lead to poor generalization and low accuracy when training the monitoring model with one tool to predict another. Fig. 17 demonstrates the comprehensive performance of the proposed model under E dataset, where each data is the average of six tests. Despite the large difference in the distribution of the training and test sets in E dataset, the model shows good generalization and robustness, especially in the most critical severe wear stage. In this stage, the precision, recall, and F1 score are 97.0 %, 98.2 %, and 97.5 %, respectively.

To further validate the model's robustness, Gaussian noise was added to the D3 dataset to simulate measurement errors in real-world environments. Samples with added noise were fed into the trained model, and the test results are shown in Fig. 18. To comprehensively evaluate the test outcomes, accuracy, macro-average precision, macro-average recall, and macro-average F1 score were adopted as evaluation metrics. The results demonstrate that even when the noise level reaches a significant 0.6, the model's metrics remain above 92.0 %. As the noise level continues to increase, the model's accuracy declines at an accelerated rate. The test data indicates that the proposed model

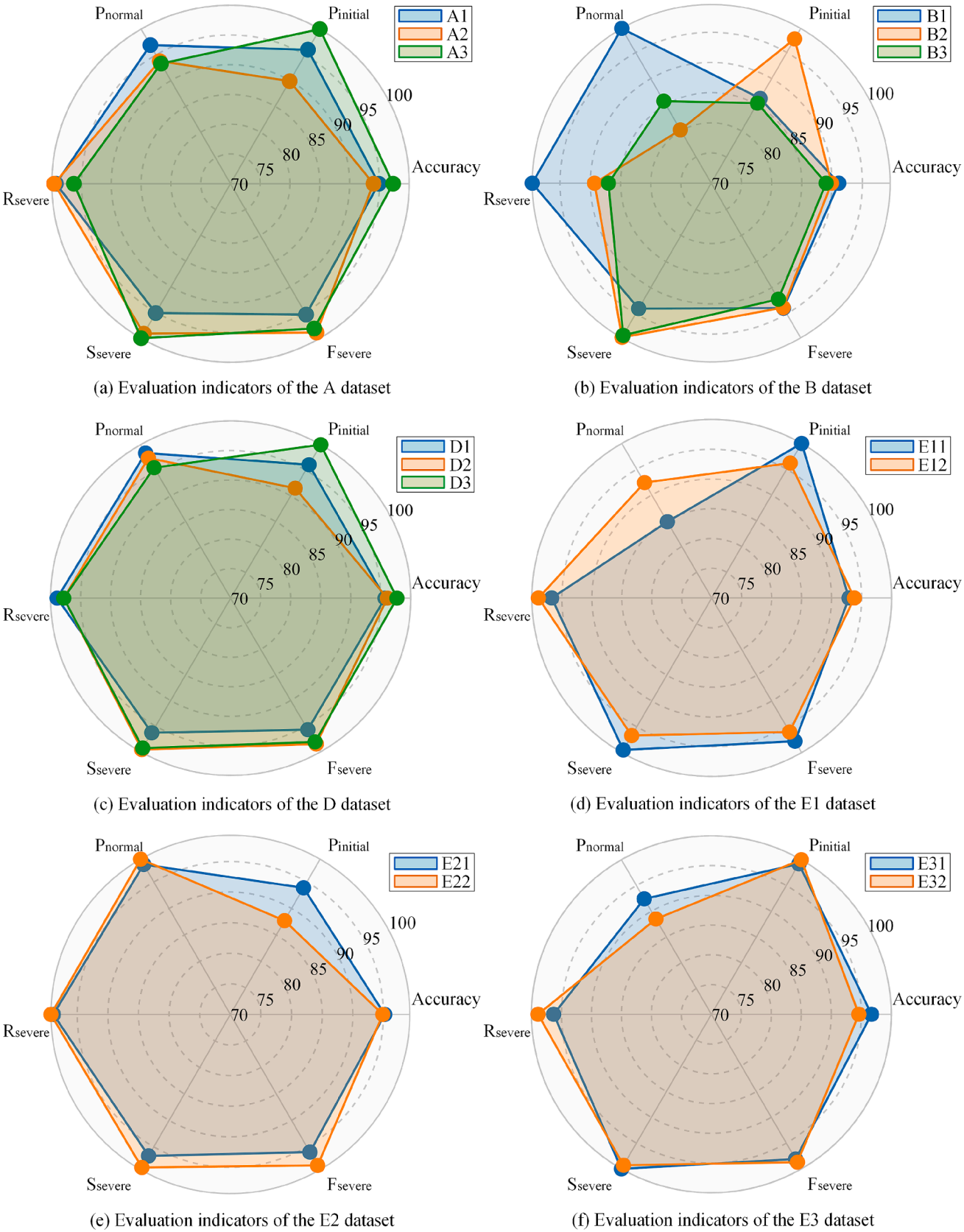


Fig. 14. Radar chart of evaluation indicators for A, B, D and E datasets.

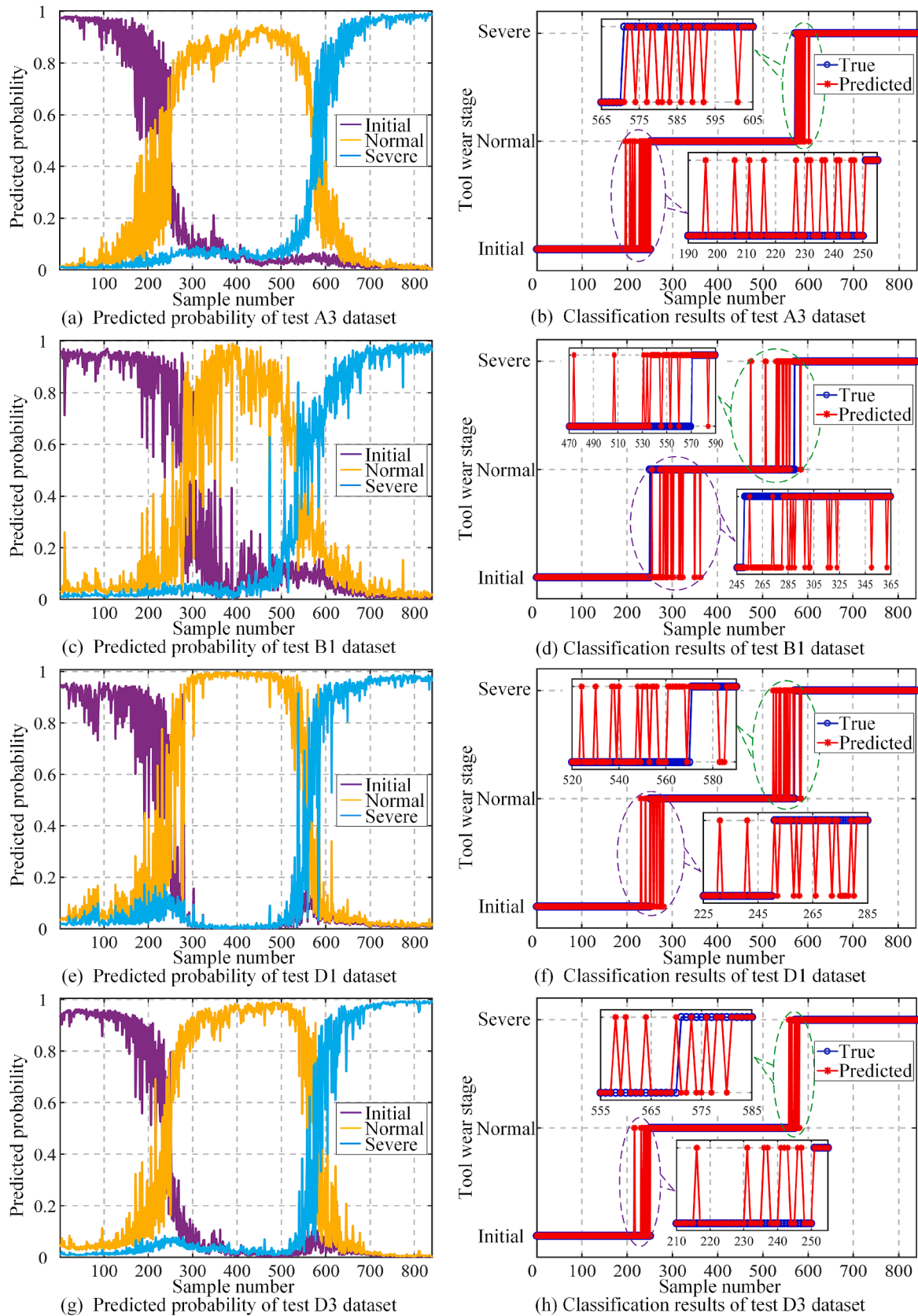


Fig. 15. Monitoring results of tool wear stages in A3, B1, D1 and D3 test sets.

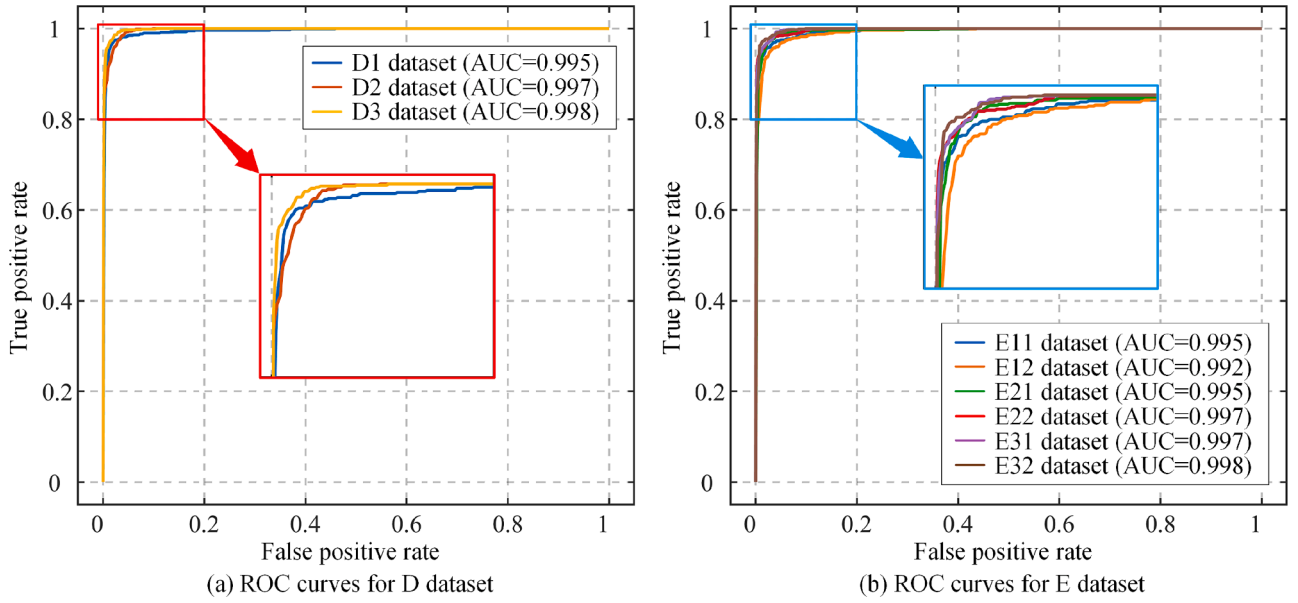


Fig. 16. ROC curves and AUC values for D and E datasets.

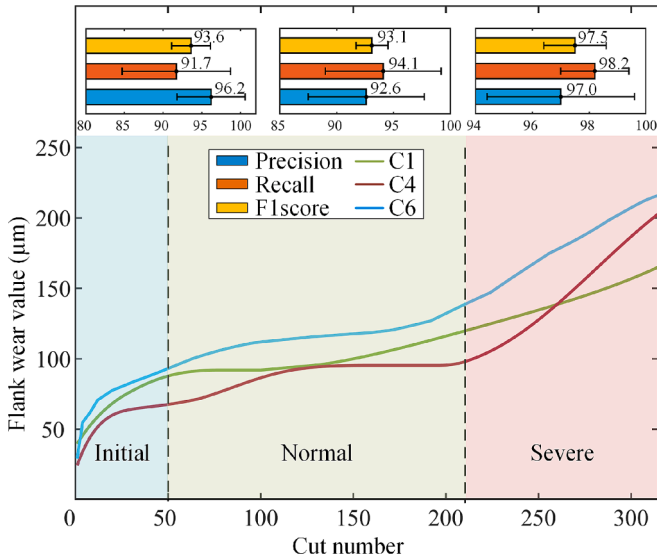


Fig. 17. Comprehensive performance of the proposed model for E dataset.

exhibits a certain degree of resistance to minor noise interference, thereby proving its robust stability.

4.3. Ablation experiment

In the ablation experiments, GNN was first used as the baseline model, and then feature extraction modules (Temporal feature extraction and spatial feature extraction) were successively stacked for comparison with the model proposed in this paper. The training and testing were performed on the D dataset. The results are shown in Fig. 19. Fig. 19(a) shows the test results of the three models on the D1 dataset, and Fig. 19(b) shows the test results of the three models on the D3 dataset. The accuracy reflects the overall performance of the model, and the F1 score is the macro average of the three wear stages, comprehensively assessing the robustness of the model. The results indicate that as the modules are stacked sequentially, the recognition performance for tool wear stages improves progressively, demonstrating that the spatio-temporal feature extraction module and the TGP based on GIN module

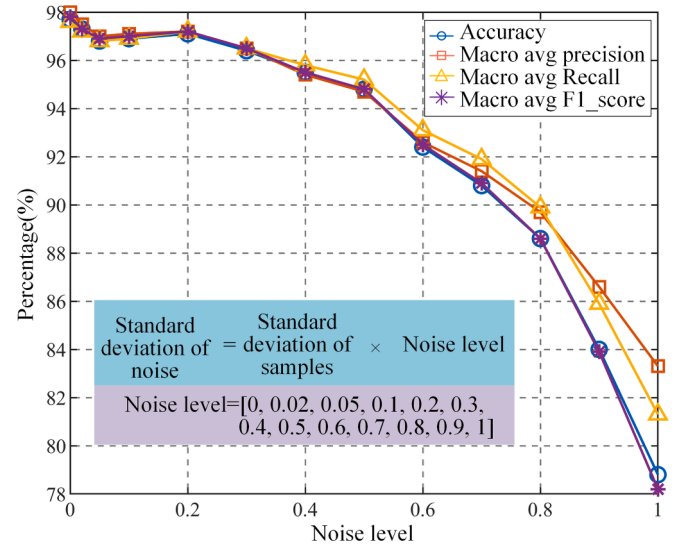


Fig. 18. The noise resistance capability of the proposed model.

proposed in the paper effectively enhance model performance.

In addition, five models were designed to verify the impact of each module on the overall model performance. The modules of temporal feature extraction, spatial feature extraction and fusion, dynamic graph generation, GIN, and TGP were removed in the order of model structure. The D dataset is utilized to evaluate the contribution of these modules to the overall model performance. The experimental results of different datasets with different models are presented in Fig. 20. From the results, it can be concluded that the average accuracy of each model is 92.8 %, 95.6 %, 94.1 %, 95.7 %, 88.4 % and 96.5 % respectively.

Additionally, the figure shows the decrease percentage in model performance after removing each module. The results show that the classification performance of the models with the elimination of the correlation module are all degraded. Model 1 shows a 3.7 % decrease in average accuracy compared to proposed model, which indicates that temporal feature extraction has a significant role in capturing temporal dependencies, and it is effective in improving the classification performance of the model. Model 3 shows a smaller decrease in performance

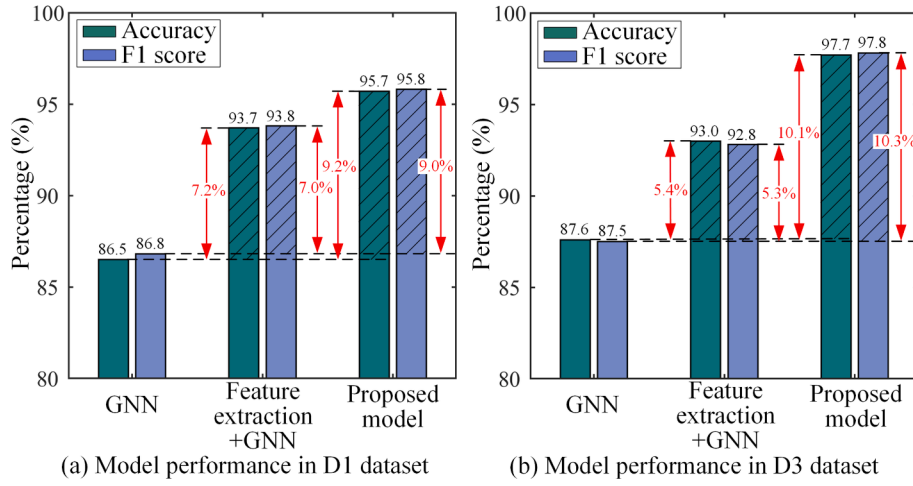


Fig. 19. Model performance in D dataset.

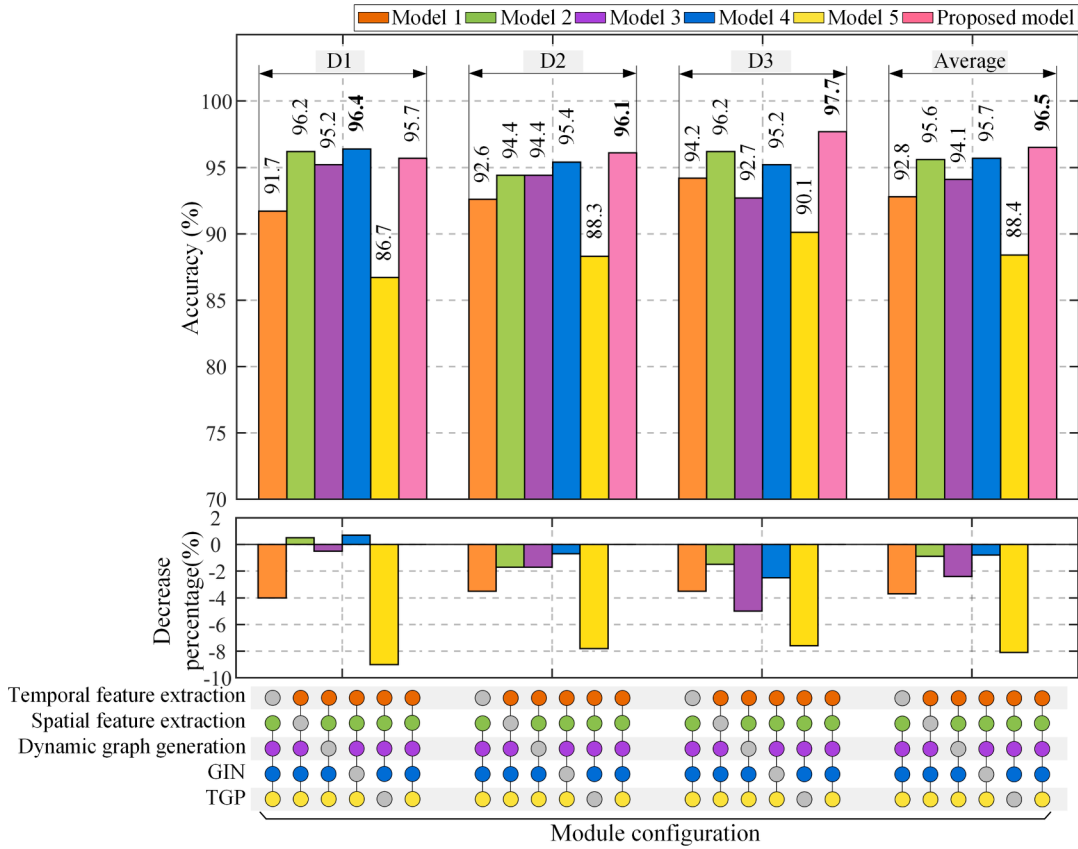


Fig. 20. Results of ablation experiments.

compared to proposed model (0.5 % for D1 and 1.7 % for D2), however, on the D3 dataset, the performance significantly decreases to 92.7 %, a decrease of 5.0 %. This indicates that the dynamic graph generation plays an important role in processing complex graph structure data, especially in scenarios where the data dynamics change significantly. Model 5 shows an average decrease of 8.1 % compared to proposed model. This suggests that the TGP plays a key role in integrating temporal dimensions and graph structure information, and its removal has the most significant impact on model performance. In addition, Model 2 and Model 4 have a role in capturing spatial structure information, although their impact on overall performance is relatively limited. Taken together, the synergy of the key components can significantly

improve the overall performance of the model, especially when dealing with complex time series and graph structure data.

4.4. Comparison with other models

In order to verify the effectiveness and superiority of the proposed model, the proposed algorithm will be compared with other published benchmark algorithms. Among these, MTF-AviTK [37], HTT-Net [47], and Transformer [48] represent applications of the Transformer model. 1DCNN-DGCCA [49] and CaAt-ResNet-1d [50] exemplify 1D-CNN. GCN [51], SGCN [52], GAT [53], and GIN [54] are fundamental GNN models. In the Table 6, the data for the first five models are directly referenced

Table 6
Comparison results between proposed model and other existing models on D dataset (%).

Models	D1	D2	D3	Average value	Standard deviation
MTF-AviTK [37]	95.38	92.81	95.71	94.63	1.30
HTT-Net [47]	94.35	93.67	98.59	95.54	2.18
GHCRCBM [55]	89.20	90.80	58.40	79.47	14.91
1DCNN-DGCCA [49]	89.50	90.00	89.20	89.57	0.33
CaAt-ResNet-1d [50]	85.25	87.98	89.27	87.50	1.68
Transformer [48]	92.02	95.60	91.07	92.90	1.95
GCN [51]	90.24	90.36	86.79	89.13	1.66
SGCN [52]	91.07	89.76	91.19	90.67	0.65
GAT [53]	90.71	90.60	91.90	91.07	0.59
GIN [54]	93.10	93.33	91.90	92.78	0.63
Proposed model	95.71	96.07	97.74	96.51	0.86

from their original papers; the latter five are constructed based on the baseline models, with three layers and trained using the parameters configured in Section 3.4. The comparison indicators use the accuracy here. On the D dataset, The accuracy of the model proposed in this paper is 95.71 %, 96.07 % and 97.74 % on the three datasets, with an average accuracy of 96.51 %, which is an overall excellent performance. Compared with existing models, the presented method performs best on the D1 and D2 datasets, and second only to HTT-Net (98.59 %) on the D3 dataset. In addition, the accuracy of the method stays above 95.71 % on all three datasets, and the standard deviation is 0.86, which is the smallest among all the models, proving that it has good stability under the premise of prioritizing accuracy. Compared with the existing methods MTF-AviTK and HTT-Net, the average accuracy of the proposed model in the paper has been improved by 1.88 % and 0.97 %, respectively, which can also prove the excellent performance of the proposed model. Furthermore, the overall strong performance of the GNN base model demonstrates the significant importance of constructing a non-Euclidean space for temporal signals and extracting features from it.

Compared to D dataset, only a limited number of researchers have utilized E dataset to validate algorithms for tool wear stage recognition. As discussed in Section 3.2 on dataset partitioning, the distributional discrepancy between the training and testing sets in E dataset is more pronounced than D dataset, thereby imposing higher demands on the generalization capability and robustness of the model. To evaluate the performance of the proposed model on E dataset, it was compared with three existing models MTF-AviTK [37], GAF-CNN [56], and HTT-Net [47], which have been tested on E dataset in prior studies. It is evident

that, except for E11 and E32 datasets, the proposed method outperforms the other models across the remaining datasets. Furthermore, the standard deviations of the three comparative models are 2.71, 3.13, and 5.43, respectively, while the proposed model exhibits the lowest standard deviation of only 1.18, demonstrating its superior generalization ability and stability compared to the other models. Fig. 21 shows the comparison of the test accuracy results of each model under E dataset.

Table 7
Cutting parameter information from the NASA ames experiment.

Case	Number of passes	Cutting depth (mm)	Feed rate (mm/r)	Workpiece material
2	14	0.75	0.5	Iron
12	15	0.75	0.5	Iron
13	15	0.75	0.25	Steel

Table 8
Division of training and test datasets.

Datasets	Training datasets	Test datasets
T1	C2 and C12	C13
T2	C12 and C13	C2
T3	C13 and C2	C12

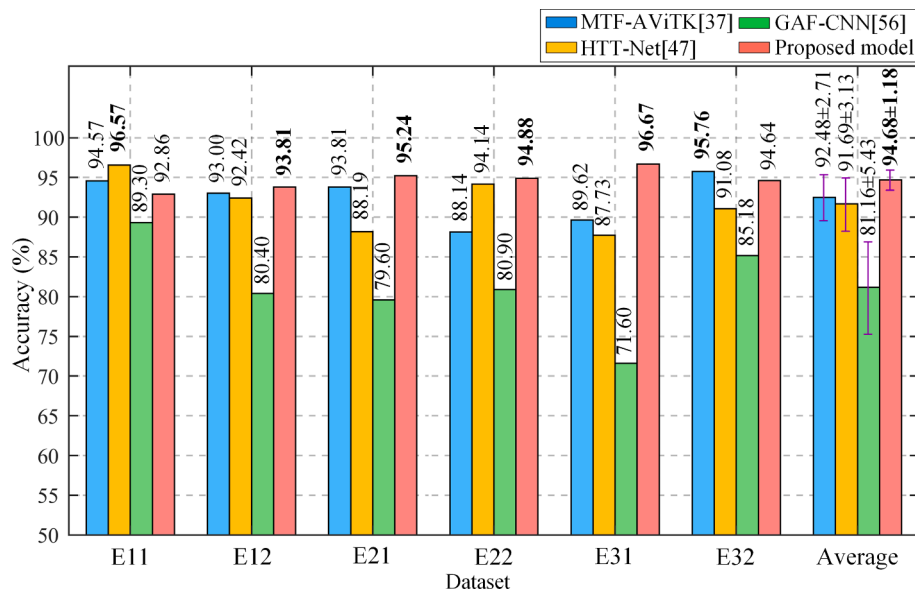


Fig. 21. Comparison results of the model accuracy on E dataset.

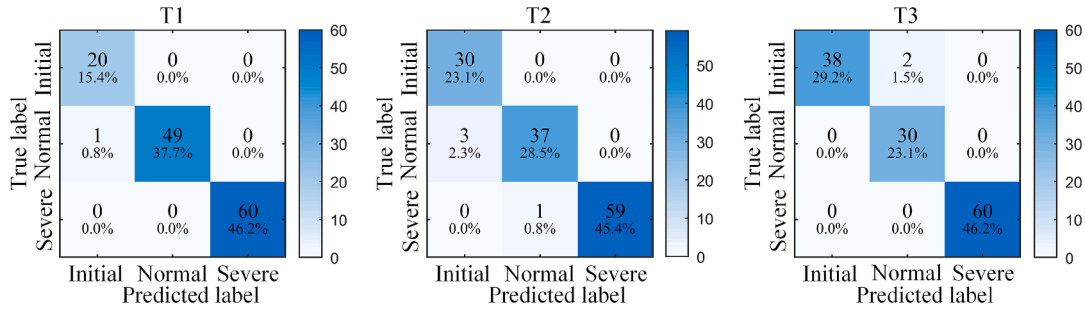


Fig. 22. The confusion matrices of T dataset.

Table 9
Evaluation indicators on the T dataset.

Indicators		T1	T2	T3
Accuracy (%)		99.2	96.9	98.5
Precision (%)	P _{initial}	95.2	90.9	100.0
	P _{normal}	100.0	97.4	93.8
Recall (%)	R _{severe}	100.0	98.3	100.0

4.5. Performance on NASA datasets

To further validate the generalization capability of this model, another publicly available dataset was employed: the NASA Ames tool wear dataset. Experiments were conducted on an MC-510 vertical machining center at a spindle speed of 826 r/min. A KC710 face milling cutter with a blade diameter of 70 mm was used to mill cast iron and stainless steel blocks measuring 483 × 178 × 51 mm. Sensors collected the following signals: spindle vibration and acoustic emission signals, worktable vibration and acoustic emission signals, as well as DC and AC signals from the spindle motor. The NASA Ames dataset involved two milling runs under each of eight distinct cutting conditions, with rake face wear values measured continuously until tool wear became severe and the tool became unusable. From the 16 cases, the following three were selected for analysis, with specific machining parameters detailed in the Table 7.

This study utilizes DC and AC signals from the spindle motor, along with acoustic emission signals from both the spindle and worktable. Data preprocessing follows Section 3.2, with samples collected from stable cutting segments. Due to the limited volume of raw data, stratified sampling was not applied to ensure sufficient training samples. Ultimately, 130 samples were collected from each case, totaling 390 samples. Three wear stages were defined based on wear volume. Three sets of cross-validation experiments were designed as Table 8.

The samples were trained and tested on the model, with the corresponding confusion matrix shown in the Fig. 22. As seen in the figure, the model performs well on the NASA dataset, correctly identifying most samples in the severe wear stage. Minor misclassifications primarily occur in the transitional zone between early wear and normal wear, due to the lack of clear distinguishing features in this area. Detailed data for each wear stage is presented in the Table 9. The accuracy rates for the three tools are 99.2 %, 96.9 %, and 98.5 %, respectively, indicating that the model correctly classifies most cases with highly stable overall performance. Furthermore, the model achieves a good balance between precision and recall, particularly excelling in handling the severe wear category.

5. Conclusions

The paper proposes an end-to-end tool wear stage recognition method based on GNN, addressing the nonlinear mapping problem between the multi-source sequential signals and tool wear states in both temporal and spatial dimensions. The effectiveness of the proposed

method is validated using publicly available datasets. The main conclusions are as follows:

- (1) A spatio-temporal correlation model for multi-source information was established, fully utilizing the complementarity of cutting force and vibration signals. Through an attention mechanism, it achieved the fusion of temporal and spatial features.
- (2) For the first time, GIN and TGP are introduced into tool wear stage monitoring. By comprehensively considering the long-term stability and short-term variability of temporal features, the time-series signal is modeled as a graph. Leveraging the inherent non-Euclidean structural relationships within the time-series signal enhances the accuracy of tool wear state recognition.
- (3) Cross-validation using the PHM2010 and NASA datasets demonstrated the model's robust overall performance, highlighting the advantages of multi-source information fusion alongside strong generalization capabilities and stability.
- (4) Experimental validation demonstrates that temporal feature extraction, dynamic graph generation, and the TGP module significantly enhance performance. Comparative analysis reveals that this model outperforms existing methods in both accuracy and stability, further validating the significant research value of extracting features from the non-Euclidean space of temporal signals.

Future research should focus on expanding dataset validation to include diverse machining processes and materials, ensuring the model's generalizability and robustness. Additionally, integrating the GNN-based approach with advanced techniques like transfer learning or reinforcement learning could enhance its adaptability to varying operational conditions. Finally, improving the model's explainability is crucial for increasing its acceptance in industrial settings.

Code availability

The codes that support the findings of this study are available from corresponding author upon reasonable request.

Consent to participate

All authors agree to participate. Consent for publication all authors agree to publish.

CRedit authorship contribution statement

Fangcun Cao: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xing Zhang:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization. **Zejiang Zhang:** Writing – review & editing, Visualization, Supervision. **Lidong Wen:** Supervision, Funding acquisition. **Tongyu Chang:** Writing – review & editing, Visualization, Supervision.

Ethics approval

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from corresponding author upon reasonable request.

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